

Lie Groups and Representations
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Disclaimer

These notes were taken during lecture using the vimtex package of the editor neovim. Any errors are mine and not the instructor's. In addition, my notes are picture-free (but will include commutative diagrams) and are a mix of my mathematical style and that of the instructor. If you find any errors, please contact me at plei@math.columbia.edu.

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Basic Notions

Definition 1.0.1. A *Lie group* is a group that is also a manifold. Here, manifold could mean a smooth manifold, complex manifold, or many other options.

Example 1.0.2. Locally, every Lie group looks like $(\mathbb{R}^n, +)$. An example of a complex Lie group is $\mathbb{C}^n$.

Remark 1.0.3. If $\mathbb{F}$ is a field with topology, then the additive or multiplicative group of $\mathbb{F}$ are topological groups, but usually not Lie groups.

Example 1.0.4. Let p be prime and consider the field $\mathbb{Q}_p$ of p -adic numbers. This has a topology, but is not locally isomorphic to a vector space. Here, the base of neighborhoods of 0 is formed by the fractional ideals $p^n \mathbb{Z}_p$, whereas neighborhoods of 0 in $\mathbb{R}^n$ are not subgroups.

Remark 1.0.5. It is possible to develop analysis for the p -adics and consider p -adic Lie groups.

More generally, one can define a class of “manifolds” by postulating local models and the corresponding algebras of functions. For real manifolds, the local model is $\mathbb{R}^n$ and the algebra is $C^\infty(\mathbb{R}^n)$. Then a map is C^∞ if and only if it pulls smooth functions back to smooth functions. To an inclusion $U'' \subset U$, we will associate a “restriction” $C^\infty(U) \rightarrow C^\infty(U'')$. This is known as a *(pre)sheaf* of algebras on M . To be a sheaf means that given the restrictions

$$C^\infty(U) \rightarrow \prod_i C^\infty(V_i) \rightrightarrows \prod_{i < j} C^\infty(V_i \cap V_j),$$

the first restriction is injective and its image is

$$\{f_i \in C^\infty(V_i) \mid \text{res}_1 f_i = \text{res}_2 f_i\}.$$

To define complex manifolds, we consider the sheaf of holomorphic functions.

For some of the “other options,” we may consider algebraic varieties over a field $\mathbb{F}$, where the algebras of functions are reduced commutative algebras of the form $\mathbb{F}[x_1, \dots, x_N]/I$. If we give up the idea of being reduced, we obtain schemes over $\mathbb{F}$. Algebraic varieties are both more flexible (singularities are allowed) and more rigid (any piece of the map determines the whole thing) than smooth manifolds.

Example 1.0.6. Consider the group $G = \text{SL}(n, \mathbb{F})$. G is a Lie group of dimension $n^2 - 1$ for $\mathbb{R}$ and $\mathbb{C}$, and in general, $\text{SL}(n, \mathbb{F})$ is the group of $\mathbb{F}$ -points in the *algebraic group* $\text{SL}(n)$ defined by the equation $\det = 1$.

In algebraic geometry, any variety contains an open set of smooth points. Because any group is a homogeneous space, all points must have the same properties, so they must all be smooth.

Definition 1.0.7. A Lie group G *acts* on a manifold M if there is a map of manifolds $G \times M \rightarrow M$ such that $1 \cdot m = m$ and $g_1 \cdot (g_2 \cdot m) = (g_1 g_2) \cdot m$.

For all $m \in M$ we can define the orbit and the stabilizer. If all of M forms one orbit, we say M is *homogeneous* or that the action is *transitive*.

Example 1.0.8. Let $M = G$. Then the action can be given by

Left $g \cdot h = gh$;

Right $g \cdot h = hg^{-1}$;

Adjoint $g \cdot h = ghg^{-1}$;

1.1 Examples of Lie Groups

For now, for real Lie groups, the notion of manifold we will use is that of a smooth manifold.

Example 1.1.1. The additive group $\mathbb{R}^n$ is a Lie group. In one dimension, the only connected manifolds are $\mathbb{R}$ and $S^1 = \mathbb{R}/\mathbb{Z} = SO(2, \mathbb{R})$.

Example 1.1.2. Recall the classification of two-dimensional manifolds. First, any Lie group is orientable, so we will consider only the orientable manifolds. These are classified by their genus, and only the torus $S^1 \times S^1$ is a Lie group. To see this, note that Lie groups have a trivial tangent bundle. A global frame for TG is given by translating a basis of $T_1 G$ by G . Therefore, we can write $TG = G \times T_1 G = G \times \mathfrak{g}$.

Remark 1.1.3. By the Hopf index theorem, the self-intersection of M inside TM is $\chi(M)$, so the index of a vector field on Σ_g is $2 - 2g$.

We will now consider some 3-manifolds. In particular, S^3 is the Lie group $SU(2)$. Note that unitary transformations of $\mathbb{C}^2$ send S^3 to itself. Therefore, we can send one basis element anywhere, but if we insist that the determinant is 1, the second basis vector is sent to a unique target. Thus $SU(2)$ acts transitively on S^3 with trivial stabilizers. Alternatively, we can write

$$SU(2) = \left\{ \begin{pmatrix} z_1 & -\bar{z}_2 \\ z_2 & \bar{z}_1 \end{pmatrix} \right\}.$$

All connected real Lie groups, as manifolds, have the form

$$G = (\text{maximal compact subgroup}) \times \mathbb{R}^r.$$

The maximal compact subgroup is a product of S^1 and simple nonabelian Lie groups. The simple nonabelian Lie groups are all built from $SU(2)$ in some sense.

Corollary 1.1.4. *The only finite-dimensional division algebras over $\mathbb{R}$ are $\mathbb{R}, \mathbb{C}, \mathbb{H}$.*

Note that $SO(4, \mathbb{R})$ acts on S^3 by rotations. Therefore we have a map $SU(2) \times SU(2) \rightarrow SO(4, \mathbb{R})$ with kernel $(-1, -1)$, so we have an exact sequence

$$1 \rightarrow (\pm 1) \rightarrow SU(2) \times SU(2) \rightarrow SO(4, \mathbb{R}) \rightarrow 1.$$

Heuristically, this means that left translation and right translation are as different as can possibly be.

Now we have seen groups like $GL(n, \mathbb{R}), SL(n, \mathbb{R}), U(n), SU(n), SO(n, \mathbb{R})$. In fact, the groups $SU(n), SO(n)$, the symplectic groups, and a few exceptional Lie groups, make up all simple compact nonabelian Lie groups (up to discrete centers).

1.2 Lie Group Actions on Manifolds

Recall the notions of action, orbit, stabilizer, etc. Then we have a map

$$G \times \{m\} \rightarrow \text{orbit}(m) \subset M$$

that is equivariant, so the differential of this map has constant rank r . Locally, the map looks like $\mathbb{R}^{n-r} \times \mathbb{R}^r \rightarrow \mathbb{R}^r$. Locally, the orbit looks like $\mathbb{R}^r$ and the stabilizer looks like $\mathbb{R}^{\dim G - r}$.

Theorem 1.2.1. *The stabilizer of any $m \in M$ is a submanifold of G . By definition, it can be upgraded to a Lie subgroup of G . In addition, there is a neighborhood U of $1 \in G$ such that $U \cdot m$ is a submanifold in M . If G is compact, then $G \cdot m$ is a submanifold.*

Remark 1.2.2. 1. $G \cdot m$ need not be a submanifold of M . The classical example is when $\mathbb{R}$ acts on M by flow along a vector field. For example, if $M = \mathbb{R}^2/\mathbb{Z}^2$, we can choose the vector field to be a constant vector field with irrational slope. The orbits are dense in M . In fact, we obtain a group homomorphism $\mathbb{R} \rightarrow \mathbb{R}^2/\mathbb{Z}^2$ with dense image.

2. For algebraic actions, orbits are much nicer because for $f : X \rightarrow Y$ algebraic, $f(X)$ contains a dense open subset of its closure and in fact is open in its closure (consider C^* acting on C).

Let G be a Lie group acting on a manifold M . We will denote the stabilizer of $x \in M$ as G_x and the orbit of x as $G \cdot x$. Note the orbit is not necessarily a submanifold, but is locally a submanifold. One such example is when $\mathbb{R}$ acts on M by time evolution according to some ODE. This type of behavior is studied in the field of dynamical systems.

Example 1.2.3. A homomorphism $G \xrightarrow{\varphi} H$ is a special case of an action. If G acts by left or right multiplication, we see that $G_{1_H} = \ker \varphi$ is a Lie subgroup of G and $\text{Im } \varphi = G \cdot 1_H$ may or may not be a Lie subgroup. We will call these “virtual Lie subgroups.”

Now we will discuss the space of orbits M/G . Right now, this is still a set, but we can canonically make it a topological space with the quotient topology.

Remark 1.2.4. This quotient space is very rarely Hausdorff, so in particular it is almost never a manifold. For example, if we consider $\mathbb{R}^\times$ acting on $\mathbb{R}$, we see that the quotient space $\mathbb{R}/\mathbb{R}^\times$ consists of two points, where the point 0 is closed and the point $\mathbb{R}^\times$ is a generic point.

Example 1.2.5. Now consider actions $\mathbb{R} \xrightarrow{\varphi} GL(2, \mathbb{R})$ acting on $\mathbb{R}^2$. Then there are several cases:

$$\varphi(t) = \begin{cases} \begin{pmatrix} e^{at} & 0 \\ 0 & e^{bt} \end{pmatrix} & ab > 0 \\ \begin{pmatrix} e^{at} & 0 \\ 0 & e^{bt} \end{pmatrix} & ab < 0 \\ \begin{pmatrix} e^{at} & 0 \\ 0 & e^{bt} \end{pmatrix} & \text{complex conjugate} \\ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \end{cases}.$$

In these cases, the orbits look like this:

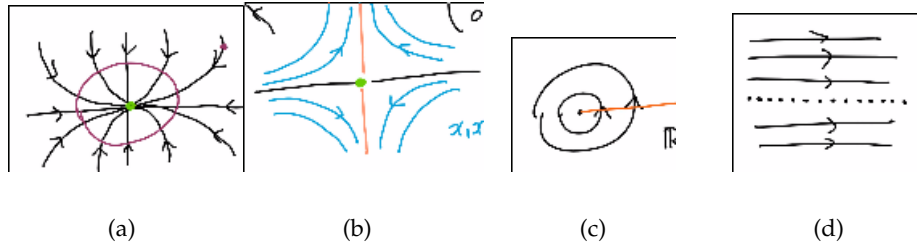


Figure 1.1: Orbits in various cases.

In the third case, the orbit space is $\mathbb{R}_{\geq 0}$. In the final case, even though the orbits are closed, the space is still not Hausdorff. In summary, M/G may be non-Hausdorff in a very complicated way.

Note that M/G has a natural sheaf of functions. If $U \subset M/G$ is open, then $\pi^{-1}(U)$ is open in M , where $\pi : M \rightarrow M/G$ is the projection. We will declare the functions on U to be the G -invariants. In the best case scenario, when U is sufficiently small, we have $\pi^{-1}(U) = U \times G$ where the action is contained entirely in the second factor. Therefore functions on U defined in the new sense are the same as normal functions on U .

However, there is no reason to expect this kind of behavior. We may get interesting behavior even for quotients by a finite group. For example, consider $M = \mathbb{R}^2$ and let $G = \{\pm 1\}$. Then M/G is simply the closure of the upper half-plane with the negative and positive real axes glued together, so we obtain a cone with total angle π . We see that $\mathbb{R}^2/\{\pm 1\}$ is a manifold near every point except 0. At 0, we study functions of the form $f(x_1, x_2)$ such that $f(-x_1, -x_2) = f(x_1, x_2)$, which are functions of $u = x_1^2, v = x_2^2, w = x_1x_2$. It is easy to see that these satisfy the equation $uv = w^2$.

Similarly, if we consider $\mathbb{Z}/m\mathbb{Z}$ acting on $\mathbb{R}^2$ by roots of unity, then we will obtain invariants $(u, v, w) = (x_1^m, x_2^m, x_1x_2)$ satisfying $uv = w^m$. This is known as the A_{m-1} surface singularity.

Remark 1.2.6. Note that $\mathbb{R}^2/\{\pm 1\}$ is very different from $\mathbb{C}/\{\pm 1\}$ because we take the second quotient in the category of complex manifolds. Here, $\mathbb{C}/\{\pm 1\} \simeq \mathbb{C}$ and the projection is a double cover branched at 0. There is a similar result for $\mathbb{C}/\zeta_m$.

More generally, we have the following result: a finite subgroup $G \subset GL(n, \mathbb{C})$ is generated by complex reflections if and only if $\mathbb{C}^n/G \simeq \mathbb{C}^n$. In general, $\mathbb{C}^n/(\text{finite subgroup of } GL(n, \mathbb{C}))$ is singular.

Example 1.2.7. Consider the permutation group $S_n \subset GL(n, \mathbb{C})$. Each transposition (ij) is a reflection in the hyperplane $x_i = x_j$. Therefore the coordinates on $\mathbb{C}^n/S_n$ are the elementary symmetric functions

$$e_k(x_1, \dots, x_n) = \sum_{1 \leq i_1 < \dots < i_k \leq n} x_{i_1} \cdots x_{i_k}$$

for $k = 1, \dots, n$. This is an important notion in representation theory because we can consider the map

$$GL(n, \mathbb{F})/\text{conjugation} \xrightarrow{\text{eigenvalues}} \mathbb{F}^n/S_n.$$

In general, S_n can be replaced by the Weyl group.

In summary, M/G may have complicated topology in singular. Complexity is good in math, but it is also good to have the simple cases. The best possible case is when the action is *proper*, i.e. that the map $G \times M \rightarrow M \times M, (g, x) \mapsto (gx, x)$ is proper (ensures the quotient is Hausdorff), and *free*, i.e. that there are no stabilizers.

Theorem 1.2.8. *Let $G \times M \rightarrow M$ be a free and proper action of a Lie group G on a manifold M . Then M/G is a manifold and the projection $M \xrightarrow{\pi} M/G$ is a locally trivial fibration with fiber G .*

Example 1.2.9. Here are some examples of a free and proper action:

1. The action of G on $H \supset G$. Then $(g, h) \mapsto (gh, h)$ is an embedding and is in particular proper. Therefore H/G is a manifold.
2. Any free action of a compact Lie group.

Proposition 1.2.10. *The map $\pi : M \rightarrow M/G$ is open.*

Proof. Note that $\pi^{-1}(\pi(V)) = \bigcup_{g \in G} g \cdot V$ is open. □

Note that the quotient by a group is a special case of a quotient by an equivalence relation.

Proposition 1.2.11. *Suppose $\pi : M \rightarrow Y$ is open and a quotient by a closed equivalent relation. Then Y is Hausdorff.*

Proof. Suppose that x, x' be such that $\pi(x, x') = (y, y')$ such that $(x, x') \notin R$. Then there exists a neighborhood of (x, x') not intersecting R , so there exist $U, U' \ni x, x'$ such that $U \times U' \cap R = \emptyset$. These project to disjoint opens, so Y is Hausdorff. □

In our situation, R is the image of $G \times M \rightarrow M \times M$. By definition, the action is proper if this map is proper.

Proposition 1.2.12. *Suppose $f : X \rightarrow Y$ is proper. Then the image of a closed set is closed.*

Proof. It suffices to prove $f(X)$ is closed. Suppose $f(x_i) \rightarrow y_\infty$. Then $x_i \in f^{-1}(\{f(x_i), y_\infty\})$, which is compact. Thus we can find a subsequence $x_{k_i} \rightarrow x_\infty \in X$, so because f is continuous, $f(x_\infty) = y_\infty$. □

In particular, in a Hausdorff space, all points are closed. In terms of our group action, this means all orbits are closed.

Proof of Theorem 1.2.8. Fix a point $x \in M$ and look at the neighborhood of $\pi(x)$. Consider the differential $\mathfrak{g} \oplus T_x M \rightarrow T_x M$ of the action map $G \times M \rightarrow M$. Because this map has maximal rank everywhere, if we choose coordinates ξ along $G \cdot x$ and η along M and ξ' for G , the differential is simply $(\xi', \xi, \eta) \mapsto (\xi' + \xi, \eta)$.

Choose a submanifold S transverse to the orbit, we can consider the action $G \times S \rightarrow M$. We see that this map is a local diffeomorphism, so we need $\pi^{-1}(\pi(S)) = G \times S$. This is not obvious and requires properness. One can imagine that there exists g such that $gS \cap S \neq \emptyset$ for any S .

Choose a sequence S_n that shrinks to x . Then we consider the set $\{g \mid gS_n \cap S_n \neq \emptyset\} \setminus \{1\}$. Then we can consider

$$\{g \mid g\bar{S}_n \cap \bar{S}_n \neq \emptyset\}$$

with a fixed neighborhood of 1 removed. This is compact by properness. If g lies in the intersection of all such sets, it must stabilize x , which is impossible. □

Remark 1.2.13. For algebraic actions, it is possible that $G \cdot x$ is free but for all open U , $\text{Stab}(x') \neq \{e\}$ for all $x' \in U$. For an example, $SL(2, \mathbb{C})$ acts on cubic polynomials in x_1, x_2 . The generic polynomial has three roots and has stabilizer μ_3 , but $x_1^2 x_2$ has trivial stabilizer.

Theorem 1.2.14. *Suppose the action of G on M is proper (but possibly not free). Then the normal bundle of $G \cdot x$ is a vector space with an action of G_x , so there exists a neighborhood of $G \cdot x$ isomorphic to $G \times N_x / G_x$. This is a vector bundle over the orbit with an action of G (it is precisely the associated bundle).*

Now note that the stabilizer of G_x is compact because the action is proper. We would like to find a G_x -invariant slice at x . To do this, we will need to discuss metrics. This is a smooth nondegenerate positive-definite quadratic form on each fiber. Then we can define the length of a curve by

$$\int \sqrt{\|\dot{x}(t)\|^2} dt.$$

There exist curves that minimize length locally, and these are called *geodesics*.

Then there is a map $T_x M \rightarrow M$ given by following a vector v along the geodesic in the direction of v for time 1. This is a local diffeomorphism and is closely related to the exponential for Lie groups. Later in the lecture, we will prove that if G acts on M and G is compact, then M has a G -invariant Riemannian metric.

In particular, there is a G_x -invariant Riemannian metric on M . Then we can write $T_x M = T_x G \cdot x \oplus (T_x G \cdot x)^\perp$. Identifying $(T_x G \cdot x)^\perp$ with the normal bundle $\nu_{G \cdot x}$, then the slice is simply $S := \exp(\nu_{G \cdot x})$.

Now consider the action $q : G \times S \rightarrow M$. If $h \in G_x$, then $q(gh, s) = q(g, hs)$ by definition. Therefore, we can write

$$q : G \times S / G_x \rightarrow M,$$

where G_x acts by $(g, x) \mapsto (gh^{-1}, hs)$. This is G -equivariant and locally an isomorphism.

Theorem 1.2.15. *This is an isomorphism of $G \times N_x / G_x \rightarrow M$ is a neighborhood of the orbit of x . Note that we can scale any $\mathbb{R}^n$ to the unit ball.*

Corollary 1.2.16. *The neighborhood of the orbit of x in M/G looks like N_x / G_x .*

Corollary 1.2.17. *The stabilizer of any nearby point is conjugate to a subgroup of G_x .*

Remark 1.2.18. Sometimes the quotient $G \times Y / H$ by the action $(g, y) \mapsto (gh^{-1}, hy)$ is denoted by $G \times_H Y$.

Corollary 1.2.19. *The manifold M has a G -invariant metric.*

Proof. Let $h \in G_x$. Then for a vector $v \in T_{g \cdot x} M$, we can associate $g^{-1}v$ and $h^{-1}g^{-1}v$ to v , but these must have the same length because they differ by an element of the stabilizer. This gives an invariant metric in a neighborhood of the orbit. Finally, we can sum the local metrics over a partition of unity to obtain a global invariant metric. $\square$

Theorem 1.2.20. *Let H be a compact Lie group acting on a manifold M . Then M has an H -invariant metric.*

Proof. Choose some Riemannian metric $\|\cdot\|_0$ on M . Then choose a Haar measure on H and define

$$\|v\|^2 := \int_H dh \|h \cdot v\|_0^2.$$

To show invariance, note that

$$\|g \cdot v\|^2 = \int_H dh \|g \cdot h \cdot v\|_0^2 = \int_H d(g^{-1}h') \|h' \cdot v\|_0^2 = \int_H dh' \|h' \cdot v\|_0^2. \quad \square$$

More generally, suppose H acts on an affine linear space by affine transformations and suppose there is a closed convex H -invariant subset B . Then there exists an H -fixed point (by the same integration argument).

Now we will show existence of the Haar measure. Recall that $TH \simeq T_1H \times H$ by left translation. Then we may choose an $\mathfrak{h}$ -valued 1-form $g^{-1}dg$ on H , and this gives a finite volume form if the group is compact.

Classification of Lie Groups

2.1 Topology of Lie Groups

Recall that if G acts on M properly and freely then $\pi : M \rightarrow M/G$ is a locally trivial fibration with fiber G and M/G is a manifold. Now suppose H is a Lie subgroup of G . Then H acts freely and properly on G . Therefore we have a map $G \rightarrow G/H$ and G/H is a manifold. Our goal is to use this fibration to understand the geometry of its ingredients.

Example 2.1.1. Consider $G = \mathrm{SU}(2) \simeq S^3$ and let H be the set of diagonal matrices in G . Then to compute G/H , note that G acts on $\mathbb{C}^2$ and hence on $\mathbb{CP}^1$. The stabilizer of a point is H , and thus $G \rightarrow G/H$ is the *Hopf fibration* $S^1 \rightarrow S^3 \rightarrow S^2$. An illustration of the Hopf fibration is below:



Figure 2.1: The Hopf fibration

Any two fibers are linked as the Hopf link, so this is not a globally trivial fibration.

Example 2.1.2. Consider $G = \mathrm{SU}(2)$ acting on itself by conjugation. This fixes $1 \in G$, so it acts by conjugation on $\mathfrak{g} = T_1 G = \{\xi \in M_2(\mathbb{C}) \mid \xi + \xi^\dagger = 0, \mathrm{tr} \xi = 0\}$. Because the norm of the matrix is preserved, we have a map $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3, \mathbb{R})$ with kernel the center of $\mathrm{SU}(2)$, which is just $\{\pm 1\}$. By dimension arguments, the map is surjective, and thus we have realized $\mathrm{SO}(3, \mathbb{R}) \simeq \mathbb{RP}^3$.

We can discuss various topological invariants of Lie groups, in particular their homotopy, homology, and cohomology groups. We will begin with $\pi_0(X)$, the set of connected components. If G is a Lie group, then $\pi_0(G)$ is a **group** isomorphic to G/G_0 , where G_0 is the connected component of the identity. It is clear that $\pi_0(G/H) = \pi_0(G)/\text{Im } \pi_0(H)$ under the natural map $\pi_0(H) \rightarrow \pi_0(G)$.

Example 2.1.3. Let $G = \text{SU}(2)$ and let $H = Z(\text{SU}(2))$. Then any path connecting ± 1 in G descends to $G/H \simeq \mathbb{RP}^3$, so the kernel of $\pi_0(H) \rightarrow \pi_0(G)$ is exactly the image of the transport $\pi_1(G/H) \rightarrow \pi_0(H)$.

Recall that we have a long exact sequence

$$\cdots \rightarrow \pi_1(H) \rightarrow \pi_1(G) \rightarrow \pi_1(G/H) \rightarrow \pi_0(H) \rightarrow \pi_0(G) \rightarrow \pi_0(G/H) \rightarrow 1$$

arising from the fibration.

Theorem 2.1.4. Let G be a topological group. Then $\pi_1(G)$ is abelian.

Proof. The homotopy $[0, 1] \times [0, 1] \rightarrow G, (t, s) \mapsto \gamma_1(t)\gamma_2(s)$ exhibits a homotopy between $\gamma_1\gamma_2$ and $\gamma_2\gamma_1$ for two loops γ_1, γ_2 based at the identity. $\square$

Remark 2.1.5. $\pi_1(\mathbb{R}^2 \setminus \pm 1)$ is not abelian. In fact it is equal to $\mathbb{Z} * \mathbb{Z} = F_2$.

Let $\tilde{X} \xrightarrow{\pi} X$ be a covering space. Then we have a map $\pi_1(X) \rightarrow \tilde{\pi} \rightarrow X$ given by transport, and if $\pi_1(\tilde{X}) = 1$, then $\tilde{X}$ is the universal cover.

Proposition 2.1.6. If G is a Lie group then so is its universal cover $\tilde{G}$ by multiplication $\gamma_1(t)\gamma_2(t)$.

Corollary 2.1.7. If G is a connected Lie group, then there exists a unique simply connected Lie group $\tilde{G}$ such that $1 \rightarrow \pi_1(G) \rightarrow \tilde{G} \rightarrow G \rightarrow 1$ is an exact sequence.

Example 2.1.8. The map $\text{SU}(2) \rightarrow \text{SO}(3, \mathbb{R})$ is a universal covering. An even more basic example is the cover $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{R} \rightarrow S^1 \rightarrow 1$.

Proposition 2.1.9. Suppose G is a connected Lie group and $\Gamma \subset G$ is a discrete normal subgroup. Then $\Gamma \subset Z(G)$.

Proof. Let $\gamma \in \Gamma$ and consider the map $G \ni g \mapsto g\gamma g^{-1} \in \Gamma$. Because G is connected, the image is a point, which must be γ because we can choose $g = 1$. $\square$

In summary, any connected Lie group G has the form $\tilde{G}/\Gamma$, where $\tilde{G}$ is simply connected and Γ is a discrete subgroup of the center.

Corollary 2.1.10. If G is abelian, then G is of the form $\mathbb{R}^n/\Lambda$, where Λ is some discrete subgroup, and this means that $G \simeq \mathbb{R}^k \times (S^1)^{n-k}$.

Remark 2.1.11. This allows us to prove the fundamental theorem of algebra. If $\mathbb{F}/\mathbb{R}$ is a field extension, then $(\mathbb{F}^*)_0$ is abelian and connected. Then for $d = 1$, this is $\mathbb{R}$, for $d = 2$ this is $S^1 \times \mathbb{R}$, and for $d \geq 3$ this is $S^{d-1} \times \mathbb{R}$, which is impossible.

Here is a very important ideal in Lie theory: Consider a Lie group G with identity 1. Then if we consider $\mathfrak{g} = T_1 G$, we can reconstruct a lot of information about G . An obvious limitation of this approach is that some Lie groups are locally isomorphic.

Example 2.1.12. Recall that $\text{SU}(2)$ is a double cover of $\text{SO}(3)$, so they are locally isomorphic. In particular, any group G is locally isomorphic to G/Γ , where $\Gamma \subset Z(G)$ is discrete.

Our strategy for dealing with this is to determine the universal cover, which is equivalent to determining $\pi_1(G)$. We will see later that simply-connected Lie groups are determined by the infinitesimal data.

Previously, we discussed the long exact sequence arising from a fibration. To make this precise, we need to define $\pi_n(X, *)$. But this is simply the group $[S^n, X]^0$ of homotopy classes of based maps from S^n to X . For $n > 1$, we can see that π_n is commutative by the following picture (or the fact that S^n is a double suspension).

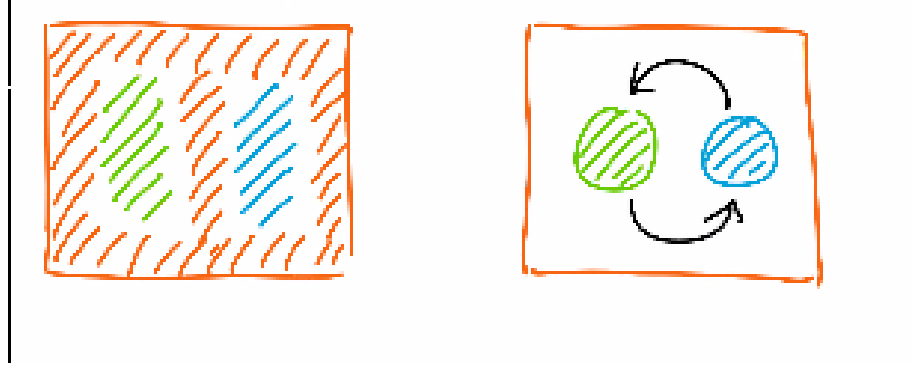


Figure 2.2: Proof that π_2 is abelian by picture

Very often, G/H is a sphere, with homotopy groups

$$\pi_k(S^n) = \begin{cases} 0 & k < n \\ \mathbb{Z} & k = n \\ ??? & k \gg n \end{cases}.$$

Not much is known about the higher homotopy groups of spheres, and computing them is a central problem in modern algebraic topology. Fortunately, it is easier to compute the homotopy groups of Lie groups.

2.2 Lie Algebras

We will discuss reconstruction of simply-connected Lie groups from their local data. This can be phrased as an equivalence of categories. On one side, we have the category of simply-connected Lie groups, and on the other side, we have the category of Lie algebras. We will define a functor Lie from Lie groups to Lie algebras that is an equivalence of categories.

We will define $\text{Lie}(G) = T_1(G)$ plus some extra data, and for any $f: G \rightarrow G'$, we will define $\text{Lie}(f) = df: T_1 G \rightarrow T_1 G'$.

Example 2.2.1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a Lie group homomorphism. Then we can differentiate to see that

$$\left. \frac{d}{dy} \right|_{y=0} f(x+y) = f'(x) = f'(0).$$

This means that the morphism is determined by an ODE, and so it must be linear. Then $f'(0)$ is precisely the map between Lie algebras.

In general for $f: G \rightarrow G'$, then we can differentiate with respect to g_2 at $g_2 = 1$ to obtain a system of first order ODEs. Then solvability of these ODEs is a condition on df that is equivalent to being a homomorphism of Lie algebras.

Definition 2.2.2. A Lie algebra is a vector space $\mathfrak{g}$ with a bilinear operation

$$[-, -]: \mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathfrak{g}$$

such that $[x, x] = 0$ that satisfies the Jacobi identity:

$$[z, [x, y]] + [y, [z, x]] + [x, [y, z]] = 0.$$

We will construct the Lie bracket out of multiplication in G . If we simply differentiate the multiplication m , the differential

$$dm: T_1 G \oplus T_1 G \rightarrow T_1 G$$

is simply the addition. This is a linear map, but it tells us that all Lie groups are the same locally.

Therefore, we need to consider higher order terms in the Taylor expansion of m . This is

$$m(\xi, \eta) = 0 + (\xi + \eta) + \text{quadratic terms} + \dots$$

and the quadratic term is a bilinear form $B(\xi, \eta)$ with no quadratic terms in ξ or η . This is **not** independent of the coordinates because if we choose $\xi' = \xi + Q(\xi)$, $\eta' = \eta + Q(\eta)$, then the multiplication becomes

$$m(\xi', \eta') = \xi + \eta + B(\xi, \eta) + Q(\xi + \eta) + \dots$$

In particular, we have $B' = B + Q(\xi + \eta) - Q(\xi) - Q(\eta)$, which does not necessarily vanish. However, it is symmetric, so we can define the *Lie bracket*

$$[\xi, \eta] = B(\xi, \eta) - B(\eta, \xi).$$

In mathematics, there is a high road and a low road. The low road is to write everything in coordinates, which is sort of what we did here. It's good to be able to take the low road, for example when we need to compute with a computer.

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Remark 2.2.3. There are many other definitions of the Lie bracket.

1. We can consider the commutator in G . If we assume $G \subset GL(n, \mathbb{F})$, then we have $[\xi, \eta] = \xi\eta - \eta\xi$. The differential of the commutator vanishes, but the second differential is precisely the bracket.

We need to prove that the commutator satisfies the Jacobi identity, which can be restated as

$$[x, [y, z]] = [[x, y], z] + [y, [x, z]].$$

Now define $\text{ad}_x(-) = [x, -]$. Then the Jacobi identity says that ad_x is a derivation of the Lie bracket.

Note that if V is a vector space with $\star: V \otimes V \rightarrow V$, then $\text{Aut}(\star) \subset GL(V)$ is an algebraic subgroup. If we apply the Lie functor, we see that

$$\text{Lie}(\text{Aut}(\star)) = \{D \in \mathfrak{gl}(V) \mid D(y \star z) = D(y) \star z + y \star D(z)\}.$$

Consider the adjoint action of G on itself. This fixes $h = 1$, so it gives a representation of G on $\mathfrak{g} = T_1 G$. If G is a matrix group, then this is really a product $\xi \mapsto g\xi g^{-1}$. If we write this out in coordinates, we see that the adjoint representation Ad of G takes values in $\text{Aut}([-,-])$ and thus when we differentiate. If we take the derivative $\text{ad} = d_1 \text{Ad}$, we obtain a map

$$g \rightarrow \text{Der}([-,-]) \subset \mathfrak{gl}(\mathfrak{g}).$$

We simply need to check that $\text{ad}_x = [x, -]$. But this is obvious because the left hand side comes from differentiating ghg^{-1} and the right hand side comes from differentiating $ghg^{-1}h^{-1}$.

Remark 2.2.4. This gives yet another definition of the Lie bracket.

2.3 Correspondence between Lie groups and Lie algebras

Now let M be a smooth manifold with an action of a Lie group G . Then $C^\infty(M)$ is an algebra that is acted on by G . Then we know that $\text{Der}(C^\infty(M))$ is simply the space of vector fields. Then we can write $f(x) = f(x_0) + df + m_x^2$, where m_x is the ideal of functions that vanish at x_0 . For any derivation, $D(m_x^2)|_{x=x_0} = 0$. This defines a tangeng vector at any $x_0 \in M$. In particular, $\mathfrak{g}$ defines C^∞ vector fields on M . Then we know that derivations form a Lie algebra.

For example, consider the action of G on itself by left translation. Then we have a map $g \rightarrow H^0(G, TG)$. In addition, it is easy to see that this gives right-invariant vector fields on G . On the other hand, right-invariant vector fields are determined by their value at 1, so we have an isomorphism $H^0(G, TG)^G \simeq \mathfrak{g}$ of vector spaces. In fact, this can be upgraded to an isomorphism of Lie algebras.

Returning to the main point, we want to prove

Theorem 2.3.1. *The functor $\text{Lie}: \{1\text{-connected Lie groups}\} \rightarrow \{\text{Lie algebras}\}$ is an equivalence of categories.*

Theorem 2.3.2. *Let G_1, G_2 be connected Lie groups with $\text{Lie}(G_i) = \mathfrak{g}_i$. Then any homomorphism $f: G_1 \rightarrow G_2$ is uniquely determined by $df: \mathfrak{g}_1 \rightarrow \mathfrak{g}_2$.*

Proof. We know that $f(g_1 g_2) = f(g_1) f(g_2)$. Then if we differentiate with respect to $g_1 = 1 + \xi$, then

$$\frac{d}{d\xi} f(g) = f(\xi) \cdot f(g).$$

This gives us a system of first order ODE. By connectedness, there is a unique solution with prescribed initial condition. $\square$

Later, we will see that the mixed partials are equal (or the curvature vanishes) if and only if df is a Lie algebra homomorphism.

Remark 2.3.3. There is no homomorphism of Lie groups from $G_1: \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R} = G_2$ corresponding to the identity homomorphism between the Lie algebras.

These ODE that we obtain from differentiating the multiplication naturally lead to the concepts of connections and curvature. Suppose we have a locally trivial fiber bundle over a base B with fibers F . Then the idea of a connection is to be able to lift paths downstairs to paths upstairs respecting concatenations of paths.

Fix a Lie group G acting on the fiber F . Then we say the *structure group* is contained in G if all transition functions may be chosen to be in G . For example, a vector bundle is a locally trivial bundle with structure group $GL(n)$. We can use the same transition functions to glue copies of G , and we obtain a principal bundle $\mathcal{P}$. Then the old bundle can be obtained using the associated

bundle construction $\mathfrak{P} \times_G F$ and a connection on a principal G -bundle induces a connection on any associated bundle.

In our case, we are talking about the trivial G -bundle over H . Then sections are maps from H to G . In coordinates, these are lifts that are invariant under the action of G on the right. This means we can consider the value at $1 \in G$, which means we have a map $\alpha: T_b B \rightarrow \mathfrak{g}$. Thus a connection can be thought of as a right-invariant Lie algebra-valued 1-form. However, this is dependent on the trivialization. If we change the section by a function $g(b)$, we know a section is constant if

$$\frac{d}{d\xi} - \alpha(\xi) = 0.$$

However, if we conjugate by g , then we need to differentiate $d(g^{-1}) = -g^{-1} dg g^{-1}$, and obtain

$$\frac{d}{d\xi} - \tilde{\alpha}(\xi) = 0,$$

where $\tilde{\alpha} = g\alpha g^{-1} + dg \cdot g^{-1}$.

Next, when does the transport along the path depend only on the endpoints? We can consider

1. Small changes, i.e. homotopies with fixed endpoints. In this case, the connection is *flat*.
2. Paths up to homotopy, i.e. $\pi_1(B)$.

Proposition 2.3.4. *A connection is flat if and only if its curvature is identically zero.*

The curvature is a certain 2-form that measures the difference between two solutions to an ODE. If we transport along $\xi_1 \xi_2 \xi_1^{-1} \xi_2^{-1}$, then we obtain the commutator

$$\left[\frac{\partial}{\partial \xi_2} - \alpha_2, \frac{\partial}{\partial \xi_1} - \alpha_1 \right] = - \left(\frac{\partial}{\partial \xi_2} \alpha_1 - \frac{\partial}{\partial \xi_1} \alpha_2 + [\alpha_2, \alpha_1] \right).$$

Thus if the connection is flat, then the curvature vanishes. In the other direction, suppose we have two homotopic paths. Then if we break down the square $[0, 1]^2$ into squares of size ε , then each square changes the result by $\varepsilon^2 \cdot \text{curvature} + O(\varepsilon^3)$, and so if we make ε small enough, the change vanishes.

Returning to our original problem, suppose we have a map $f: H \rightarrow G$. Then $df: \mathfrak{h} \rightarrow \mathfrak{g}$ determines a connection on $G \times_H H$. On H we have the canonical 1-form $dh \cdot h^{-1}$. If df is a Lie algebra homomorphism, then

$$[df(\xi_1), df(\xi_2)] = df([\xi_1, \xi_2])$$

and thus the curvature of the connection α induced from $dh \cdot h^{-1}$ is the differential of the curvature of $dh \cdot h^{-1}$, which is identically zero.

Remark 2.3.5. All of this can be expressed in elementary terms. First, we have $\frac{\partial}{\partial \xi_i} g = \alpha_i g$. Then we have

$$0 = \frac{\partial^2 g}{\partial \xi_i \partial \xi_j} - \frac{\partial^2 g}{\partial \xi_j \partial \xi_i} = \left(\frac{\partial \alpha_j}{\partial \xi_i} - \frac{\partial \alpha_i}{\partial \xi_j} + [\alpha_i, \alpha_j] \right) g.$$

Thus we have proved the following theorem:

Theorem 2.3.6. *If H is a simply connected Lie group and $\varphi: \text{Lie } H \rightarrow \text{Lie } G$ is a Lie algebra homomorphism, then there is a unique map $f: H \rightarrow G$ such that $df|_1 = \varphi$.*

For example, if we want to prove that $\log(xy) = \log x + \log y$, we write

$$\log(xy) = \int_1^x \frac{dt}{t} + \int_x^{xy} \frac{dt}{t}$$

and note that the second term in the sum equals $\int_1^y \frac{dt}{t}$.

Recall the differential equations that we constructed for a homomorphism of Lie groups from the Lie algebra. These equations are right-invariant in both the source and the target and imply that f is a homomorphism.

Example 2.3.7. Here is a silly example: Note the isomorphism $(\mathbb{R}_{>0}, \times) \rightarrow (\mathbb{R}, +)$. Then $f(xy) = f(x) + f(y)$ and so we have

$$y \frac{d}{dy} f = c$$

for some constant c . In the other direction, we have $\varphi(x+y) = \varphi(x)\varphi(y)$, so

$$\frac{d}{dy} \varphi = c \cdot \varphi$$

for some constant c . The solution to the second equation is clearly the exponential function, and the solution to the first is

$$f(y) = \int_1^y c \frac{dt}{t} = c \log y.$$

Invariance implies that $\log$ is a homomorphism.

Theorem 2.3.8 (Lie). *For any simply connected Lie group G , the map*

$$\text{Hom}_{\text{Lie Groups}}(H, G) \xrightarrow{d} \text{Hom}_{\text{Lie Algebras}}(\text{Lie}(H), \text{Lie}(G))$$

is an isomorphism.

Here are some applications:

1. Any connected abelian Lie group G of dimension n is of the form $\mathbb{R}^n/\Gamma$, where $\Gamma \cong \mathbb{Z}^k$. Thus $G = (S^1)^k \times \mathbb{R}^{n-k}$.
2. Not all Lie groups are matrix Lie groups. However, every Lie algebra is a matrix Lie algebra in characteristic 0. In the category of Lie algebras, we can always lift the adjoint representation $\text{ad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ to the center of $\mathfrak{g}$. However, when we consider the adjoint representation of G as a Lie group, we cannot lift to the center.

For example, consider $\text{SL}(2, \mathbb{R})$. This has $\pi_1(\text{SL}(2, \mathbb{R})) = \mathbb{Z}$. Then $G = \widetilde{\text{SL}(2, \mathbb{R})}$ has $\mathbb{Z}$ in the center, and so any map

$$f: G \rightarrow \text{GL}(N, \mathbb{C})$$

corresponds to a map of Lie algebras $\mathfrak{sl}(2, \mathbb{R}) \rightarrow \mathfrak{gl}(N, \mathbb{C})$. This gives a map $\mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{gl}(N, \mathbb{C})$, which then lifts to a map $\text{SL}(2, \mathbb{C}) \rightarrow \text{GL}(N, \mathbb{C})$. In particular, all linear representations of G factor through $\text{SL}(2, \mathbb{C})$ and are thus trivial on the center.

Definition 2.3.9. Suppose G is a real Lie group with Lie algebra $\mathfrak{g}$. Then if we take the complexification $\mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$, this gives a map $G \rightarrow G_{\mathbb{C}}$ for $G_{\mathbb{C}}$ the complex Lie group corresponding to $\mathfrak{g}_{\mathbb{C}}$. We say that $G_{\mathbb{C}}$ is a *complexification* of G and that G is a *real form* of $G_{\mathbb{C}}$.

Remark 2.3.10. Note that this relation is generally many-to-one. In particular, both $SL(n, \mathbb{R})$ and $SU(n)$ are real forms of $SL(n, \mathbb{C})$.

Now let G be a Lie group and $\xi \in \mathfrak{g}$. Then $\mathbb{R} \ni t \mapsto t\xi \in \mathfrak{g}$ is a homomorphism of Lie algebras. Thus there exists a unique map

$$\mathbb{R} \ni t \mapsto \exp(t\xi) \in G$$

and this is the matrix exponential for matrix groups. Also, $\frac{d}{dt} \exp(t\xi) = \xi \exp(t\xi)$. In addition, if $[\xi, \eta] = 0$, then we can exponentiate $\exp(t\xi + s\eta) = \exp(t\xi) \exp(s\eta)$ in either order.

Theorem 2.3.11 (Lie). *For any Lie algebra $\mathfrak{g}$ over $\mathbb{R}$ or $\mathbb{C}$, there exists a unique simply-connected Lie group G with Lie algebra $\mathfrak{g}$.*

This gives a correspondence between our linear local data and nonlinear global data. We can construct manifolds either as:

1. A quotient of something simpler. If G is a simply-connected Lie group, then if we choose a point $x \in \mathfrak{g}$, then we can consider smooth paths g from 1 to x . Then $\dot{g}(t)g^{-1}(t) = \xi(t)$ the tangent vector at time t , and for a smooth homotopy between paths, If the curvature vanishes, then we have

$$\frac{\partial}{\partial s} \xi - \frac{\partial}{\partial t} \eta = [\xi, \eta].$$

Then we can write G as the paths in the Lie algebra modulo solutions to the equation. Fortunately, the analysis reduces to first-order deformations.

2. As a “submanifold” of something simpler. Write $\mathfrak{g} \hookrightarrow \mathfrak{gl}(n, \mathbb{R})$. Then we have a map $G \rightarrow GL(n, \mathbb{R})$ with some kernel Γ . Thus G is the universal cover of $G/\Gamma \subset GL(n, \mathbb{R})$. Therefore, at least locally, every element of the Lie group is a matrix.

A problem with this approach is that G/Γ need not be a submanifold. If we have the map $\mathbb{R} \rightarrow \mathbb{R}^2/\mathbb{Z}^2$ with dense image, we obtain a foliation, so individual leaves are (locally) submanifolds, but we do not globally obtain a submanifold. In particular, if $H \rightarrow G$ is an injective Lie algebra homomorphism, we have a foliation with leaves corresponding to the cosets of H in G . In particular, we have the field of tangent planes $f(\text{Lie } H) \cdot g$. Thus the cosets may be reconstructed either as leaves of the foliation or as integral manifolds for this field of tangent ($k = \dim H$)-planes (a section of a bundle of Grassmannians over M).

Note that an *integral manifold* for a field of tangent k -planes is a k -dimensional manifold $L \xrightarrow{\iota} M$ which is locally a submanifold such that $T_x L$ is precisely the value of the field of k -planes at every point $x \in L$. The idea to construct this in our situation is to start by finding an embedding $\mathfrak{g} \hookrightarrow \mathfrak{gl}(N, \mathbb{C})$. The main obstacle to this plan is that a field of k -planes may not have any integral manifolds for $k > 1$.

A connection is a special case of a field of k -planes. Here, a connection on a locally trivial bundle Π gives a field of tangent planes that are transverse to the fibers of Π . Then the existence of integral manifolds is equivalence to flatness. There is a classical criterion for the existence of integral manifolds.

Theorem 2.3.12 (Frobenius). *A field V of tangent k -planes in M has integral manifolds if and only if for all $m \in M$ the set of vector fields tangent to V forms a Lie subalgebra of $\Gamma(M, TM)$.*

Proof. Suppose we have integral manifolds $f_1 = c_1, \dots, f_{n-k} = c_{n-k}$. Then a vector field v is tangent to the integral manifolds if and only if $\frac{d}{dv} f_i = 0$, which implies that $\left[\frac{d}{dv_1}, \frac{d}{dv_2} \right] f_i = 0$.

Conversely, suppose that $\mathcal{V}$ is our field of k -planes. Then denote $\Gamma_{\mathcal{V}}(M)$ to be the set of vector fields tangent to $\mathcal{V}$. Then we have this following picture:

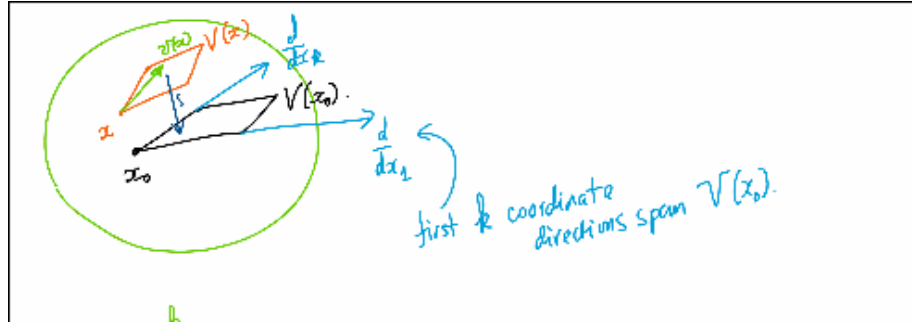


Figure 2.3: Projection of tangent planes

Thus we have

$$v = \sum_{i=1}^k c_i(x) \frac{d}{dx_i} + \sum_{i=k+1}^n c_j(x) \frac{d}{dx_i}.$$

Here the first c_i are arbitrary and the last c_j are uniquely determined. In particular, we have a basis of the form

$$v_i = \frac{d}{dx_i} + \sum_{j>k} c_j(x) \frac{d}{dx_j}.$$

Then we see that

$$[v_i, v_j] = 0 + \sum_{j>k} c_j \frac{d}{dx_j}$$

and so $[v_i, v_j] = 0$. This means that locally, the connection has curvature zero, and thus there are integral manifolds. $\square$

Returning to our case, consider $\mathrm{GL}(N, \mathbb{C})$ and consider the tangent field $\mathfrak{g}g$, where $\mathfrak{g} \subset \mathfrak{gl}(N, \mathbb{C})$ is a Lie subalgebra. Then

$$\Gamma_V = C^\infty(G \otimes) \{ \xi g \mid \xi \in \mathfrak{g} \}.$$

The right-hand factor is already closed under the commutator. By the Leibniz rule, we see that Γ_γ is also closed under $[-, -]$ and hence has integral manifolds.

Choosing the integral manifold that contains $1 \in \text{GL}(N, \mathbb{C})$, this is a subgroup of $\text{GL}(N, \mathbb{C})$. However, it is not a Lie subgroup in general. For any integral manifold L and $h \in \text{GL}(N, \mathbb{C})$, then Lh is also an integral manifold because the field was right invariant. If $h \in G$, then $Gh^{-1} = G$ because Gh^{-1} is integral and contains 1, so for all $g_1, g_2 \in G$, then $g_1g_2^{-1} \in G$. Hence G is a connected Lie group. If G is not 1-connected, then we can take the universal cover. Thus we have proved

Theorem 2.3.13 (Lie). *For any Lie algebra $\mathfrak{g}$ over $\mathbb{R}$, there exists a unique simply-connected Lie group G with $\text{Lie}(G) = \mathfrak{g}$.*

Remark 2.3.14. When is $\mathfrak{g} \subset \mathrm{GL}(N, \mathbb{C})$ algebraic? Not every Lie algebra over $\mathbb{C}$ is a Lie algebra of an algebraic group. Most differential equations with algebraic coefficients do not have algebraic solutions.

For example, the irrational winding of the torus corresponds to $\left\{ \frac{\log z}{\log w} = \text{const} \right\} \subset \mathbb{C}^* \times \mathbb{C}^*$.

Representations

Let G be a group. Then a *representation* of G over a field $\mathbb{F}$ is a map $G \rightarrow GL(n, \mathbb{F})$. Here, we will take $\mathbb{F} = \mathbb{R}, \mathbb{C}$ and strongly prefer $\mathbb{F} = \mathbb{C}$. We will call a representation of G a G -module. These form an abelian category, which is not something we will dwell on too much. A map between two representations is an “intertwining operator” which is something that makes

$$\begin{array}{ccc} V_1 & \xrightarrow{f} & V_2 \\ \downarrow g & & \downarrow g \\ V_1 & \xrightarrow{f} & V_2 \end{array}$$

commute. Then both $\ker f, \text{Im } f$ are submodules. If V has a nontrivial submodule, then it is *reducible*. Otherwise, we call it irreducible. If we have an exact sequence

$$0 \rightarrow V_1 \rightarrow V \rightarrow V_2 \rightarrow 0,$$

then V_1 is a submodule of V and V_2 is a quotient.

Definition 3.0.1. A representation V is called *semisimple* (or completely reducible), if $V = \bigoplus V_i$, where V_i is irreducible.

If V_1, V_2 are representations, then G acts on $V_1 \otimes V_2$ by $\pi_1(g) \otimes \pi_2(g)$. This comes from the map $\Delta: G \rightarrow G \times G$.

We have a notion of *characters* that send a representation V to the conjugation-invariant function $\chi_V(g) = \text{tr}_V g$. Then it is easy to see that $\chi_{V_1 \oplus V_2} = \chi_{V_1} + \chi_{V_2}$ and that $\chi_{V_1 \otimes V_2} = \chi_{V_1} \cdot \chi_{V_2}$. This is a semiring homomorphism, so we can form the *representation ring* Rep_G . This is the K -group of the category Mod_G . Also, note that if

$$0 \rightarrow V_1 \rightarrow V \rightarrow V_2 \rightarrow 0$$

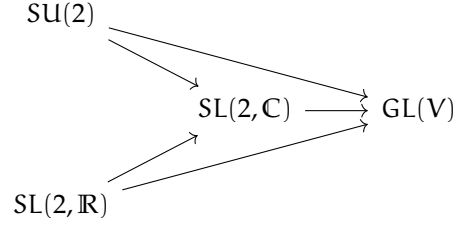
is exact, then we see that $\chi_V = \chi_{V_1} + \chi_{V_2}$. Thus we can impose the relation $[V] = [V_1] + [V_2]$.

At first sight, it seems that taking the character loses a lot of information. However, if we have all of the traces, this means we can compute all of the eigenvalues. Thus, we can reconstruct V up to conjugation from its character.

Next, if V is a G -module, then G also acts on V^* by $(g\ell)(v) = \ell(g^{-1}v)$. Also, $(V^*)^* = V$ as a G -module.

3.1 Finite Dimensional Representations

Consider the simplest groups we know: $SU(2)$, $SL(2, \mathbb{R})$, $SL(2, \mathbb{C})$. First, all of these groups have the same finite-dimensional representations given by



and in the other direction, we can complexify $\mathfrak{su}(2)$ to $\mathfrak{sl}(2, \mathbb{C})$, and then $SL(2, \mathbb{C})$ is simply-connected. However, note that $GL(1, \mathbb{C})$ is not simply-connected, so we cannot use the same argument for $U(1)$. Also, representations of S^1 and $GL(1, \mathbb{C})$ are semisimple, but $\mathbb{R}$ has the representation

$$z \mapsto \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix},$$

which is a nontrivial representation in two dimensions that is not semisimple.

Second, we note that all representations are semisimple because $SU(2)$ is compact. To see this, note that $SU(2)$ is compact and thus every representation has an invariant Hermitian form (by averaging). Then for $V_1 \subset V$, we can write $V = V_1 \oplus V_1^\perp$.

Then all irreducible representations of $SL(2, \mathbb{C})$ can be described as symmetric powers $S^k \mathbb{C}^2 \cong \mathbb{C}[(\mathbb{C}^2)^*]_k$. This has basis $v_1^{k_1} v_2^{k_2}$, where $k_1 + k_2 = k$. Here, the maximal torus acts by

$$\begin{pmatrix} z & \\ & z^{-1} \end{pmatrix} \mapsto \begin{pmatrix} z^k & & \\ & z^{k-2} & \\ & & \ddots & \\ & & & z^{-k} \end{pmatrix}.$$

We will find this structure in the representation of $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{C})$. We know $\mathfrak{g}$ has a basis

$$h = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}, e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then we see that

$$h \mapsto \begin{pmatrix} k & & \\ & k-2 & \\ & & \ddots & \\ & & & -k \end{pmatrix}$$

and $e = v_1 \frac{\partial}{\partial v_2}$, $f = v_2 \frac{\partial}{\partial v_1}$. Also, note that $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. Then e shifts the weight by $+2$ and f by -2 .

Lemma 3.1.1. *If v is an eigenvector of h with eigenvalue λ , then $h(ev) = (\lambda + 2)ev$ and $h(fv) = (\lambda - 2)fv$.*

Proof. Note that $h(ev) = [h, e]v + ehv = 2ev + e\lambda v = (\lambda + 2)ev$. A similar argument gives the result for f . $\square$

Classification of irreps of $\mathfrak{sl}(2)$. Let V be irreducible and v be an eigenvector of h with eigenvalue λ . If $ev \neq 0$, then replace v by ev . The eigenvalue cannot grow forever, so eventually we reach v such that $hv = \lambda v$ and $ev = 0$. This is called the highest weight vector.

Now we will show that V is the span of $v, fv, f^2v, \dots$. This is clearly invariant under h, f by construction. Then note that

$$ef^m v = [e, f^m]v + f^m ev = [e, f^m]v.$$

Because $[e, f^m]v$ is a combination of h, f by the commutation relations, we see that the span of $v, fv, \dots$ is a subrepresentation, so it must be everything. Then because $h \cdot f^m v = (\lambda - 2m)f^m v$, then there exists a minimal m such that $f^m v = 0$. This implies that $\lambda = m - 1$, but to show this, consider $ef^m v$, which is a multiple of $f^{m-1}v$.

The rest of this proof is left as an exercise. $\square$

Remark 3.1.2. Representations of $GL(n, \mathbb{C})$ correspond to integers $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$.

3.2 Harmonic Analysis on Compact Groups

Let G be a compact Lie group and consider representations $G \rightarrow GL(V)$. We would like to do harmonic analysis on G . Our prototype will be $G = \mathbb{R}/\mathbb{Z}$. If dx is the Haar measure, then we can write

$$L^2(G, dx) = \widehat{\bigoplus_{n \in \mathbb{Z}} \mathbb{C} \cdot e^{2\pi i n x}}$$

as a **Hilbert Space**. Recall that a Hilbert space is a complete inner product space, and the inner product on L^2 is

$$(f, g) = \int_X f \bar{g} dx.$$

We can write $\|f\|^2 = (f, f)$. Then recall that $x \mapsto e^{2\pi i n x}$ are precisely the irreducible representations of G . Here, $\widehat{\bigoplus}$ is the closure of the algebraic direct sum. Next, for any $f(x) \in L^2$, we can take the Fourier transform

$$f(x) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{2\pi i n x},$$

where $\hat{f}(n) = (f, e^{2\pi i n x})$. We will generalize this to an arbitrary compact group.

Our main issue is that for non-abelian groups, not all irreducible representations have dimension 1. However, we will have a correspondence

$$\left(\begin{array}{c} \text{matrix elements of} \\ \text{irreducible representations} \end{array} \right) \longrightarrow L^2(G, d\mu),$$

where μ is the Haar measure. These matrix elements of irreducible representations are in fact analytic. When we pass to $G_{\mathbb{C}}$, they become holomorphic.

Then if V is a complex representation of G , then its matrix elements are a map $V^* \otimes V \rightarrow C^\infty(G)$, where

$$(\ell \otimes v)(g) = \ell(gv).$$

These satisfy the following orthogonality relation which compares the inner product in $L^2(G)$ and on $\bigoplus V_i^* \otimes V_i$.

Proposition 3.2.1. *Every irreducible representation V has a unique G -invariant Hermitian inner product up to multiple.*

Proof. To show existence, take any Hermitian inner product and average over the group G . Then if $(-, -), \langle -, - \rangle$ are different invariant inner products, we can write

$$\langle v_1, v_2 \rangle = (Bv_1, v_2)$$

for some Hermitian matrix B . But then this commutes with G , so by Schur's Lemma, B is a constant. $\square$

Lemma 3.2.2 (Schur). *Let V_1, V_2 be irreducible G -modules. Then*

$$\text{Hom}_G(V_1, V_2) = \begin{cases} 0 & v_1 \not\simeq v_2 \\ \mathbb{C} & v_1 \simeq v_2 \end{cases}.$$

Proof. If $f \in \text{Hom}_G(V_1, V_2)$, then $\ker f \subset V_1$ and $\text{Im } f \subset V_2$. Thus either the kernel is everything and the image is zero, or the image is everything and the kernel is zero, so either f is zero or an isomorphism.

Now assume $V_1 \simeq V_2$ and consider $\text{Hom}_G(V, V)$. Then $\text{Hom}_G(V, V)$ is a division algebra over $\mathbb{C}$. But over $\mathbb{C}$, this must be $\mathbb{C}$ because for λ an eigenvalue, then $f - \lambda$ has nontrivial kernel and thus must be the zero map. $\square$

Now for the irrep V , take the unique inner product $(-, -)$. This gives $V^* \otimes V$ a canonical inner product. To write this concretely, write $V^* \otimes V = \text{End } V$, and then write $(A, B) = \text{tr } AB^\dagger$. For matrices E_{ij}, E_{kl} , we have

$$(E_{ij}, E_{kl}) = \delta_{ik} \delta_{jl}.$$

Theorem 3.2.3 (Orthogonality). *Let $V_1, V_2, \dots$ be a collection of distinct irreducible representations. Then consider the space $\bigoplus_i V_i^* \otimes V_i$ with inner product $(A, B) = \frac{1}{\dim V_i} \text{tr } AB^\dagger$. Then the embedding*

$$\bigoplus_i V_i^* \otimes V_i \xrightarrow{\text{matrix elements}} L^2(G)$$

is an isometry.

Proof. Denote $g \xrightarrow{\pi_i} GL(V_i)$. Then take an arbitrary $f: V_i \rightarrow V_j$ and make it G -invariant by averaging

$$\bar{f} = \int_G dg \pi_j(g) f \pi_i(g)^{-1}.$$

By Schur, we see that

$$\bar{f} = \begin{cases} 0 & i \neq j \\ \frac{\text{tr } f}{\dim V_i} & i = j \end{cases}.$$

Because the inner product on each V_i is G -invariant, then $G \rightarrow U(V_i) \subset GL(V_i)$. Therefore, for all $f \in \text{Hom}(V_i, V_j)$,

$$\int_G dg \pi_j(g) f \pi_i(g)^* = \begin{cases} 0 & i \neq j \\ \frac{\text{tr } f}{\dim V_i} & i = j \end{cases}.$$

This equality of matrices is equivalent to the desired result. $\square$

Theorem 3.2.4 (Peter-Weyl). *Let G be compact. Then if μ is the Haar measure, we have*

$$L^2(G, d\mu) = \widehat{\bigoplus_{\text{irreps } V} V^* \otimes V}$$

as modules over $G \times G$.

The key statement is that $\bigoplus V^* \otimes V$ is dense in $L^2(G)$. For the second part, note that $G \times G$ acts on $L^2(G)$ by

$$[(g_1, g_2) \cdot f](h) = f(g_1^{-1} h g_2).$$

Then $\varphi_{\ell v}(h) = \ell(h \cdot v)$, and under (g_1, g_2) , this becomes $\ell(g_1^{-1} h g_2 v) = \varphi_{g_1 \ell, g_2 v}(h)$, so $\bigoplus V_i^* \otimes V_i \rightarrow L^2(G)$ is a map of $G \times G$ modules.

Conversely, the subspace of $L^2(G)$ that transforms in V under the right regular action of G is $V^* \otimes V$. Indeed, suppose $f_k(h)$ are such that

$$f_k(hg) = \sum_j \pi_{k\ell}(g) f_\ell(h).$$

In particular, if we take $h = 1$, we get $f_k \in V^* \otimes V$.

Here are some reformulations of Peter-Weyl:

1. The unitary representation

$$\left(\bigoplus V^* \boxtimes V \right)^\perp$$

cannot have finite dimensional submodules, so we have a matrix element of an infinite-dimensional unitary representation of G . Thus Peter-Weyl is equivalent to every irreducible representation of G being finite-dimensional.

2. Every compact group has a faithful finite-dimensional representation $G \hookrightarrow GL(n, \mathbb{C})$ for some n . To see this, suppose $G \subset GL(V)$. Then $G \subset U(V)$. We can now consider the algebra generated by matrix elements g_{ij} , where $g_{ij} g_{k\ell}$ is a matrix element of $V \otimes V$. Then we see that

$$\bigoplus_k (V^{\otimes k})^* \otimes V^{\otimes k} \longrightarrow \left(\begin{array}{c} \text{matrix elements of} \\ V \otimes \dots \otimes V \end{array} \right) \subset L^2(G).$$

This is an algebra of complex valued functions stable under complex conjugation, which separates points of G . By Stone-Weierstrass, this is dense in $C(G)$ and thus in $L^2(G)$.

Remark 3.2.5. Let V be a faithful finite-dimensional representation of a compact group G . Then any irreducible representation of G is contained in the decomposition of $V^k \otimes (V^*)^\ell$. However, for $U(n)$, the defining representation is not in the decomposition of $(\mathbb{C}^n)^{\otimes k}$.

Remark 3.2.6. This proves Peter-Weyl for all compact groups because they are all matrix groups. Also, we showed that $L^2(G)$ is separable, which means that compact groups have only countable many irreps.

Continuing the proof of the second item, we have an exact sequence

$$1 \rightarrow G_N \rightarrow G \rightarrow GL \left(\bigoplus_{i \leq N} V_i \right).$$

Then $G_1 \supset G_2 \supset \cdots$ has to eventually stabilize and write G_∞ for the colimit. Then if $G_\infty = 1$, we are done. Otherwise, we have a contradiction because all functions take the same value on G_∞ -cosets.

Now we will continue our proof of Peter-Weyl. Our strategy is to break $L^2(G)$ into finite-dimensional G -invariant pieces. We can consider the $G \times G$ invariant metric on G and consider the corresponding Laplace operator and its eigenspaces. Note that $\mathfrak{g}$ has a positive-definite invariant metric and an invariant tensor in $S^2\mathfrak{g}^* \rightarrow S^2\mathfrak{g}$. Then if ξ_i is an orthonormal basis of $\mathfrak{g}$, then $\mathfrak{d}_i$ is a first-order differential operator of G , and we define the Laplacian to be

$$\Delta = \sum \xi_i^2,$$

which is also called the “Casimir¹ element.” Because this is invariant, it acts by a scalar in V , which is the eigenvalue of the Laplacian in $V^* \boxtimes V$.

Instead of doing this, we will use integral operators because they are easier to work with. We have an action of G on $L^2(G)$ on the right, which yields $f(h) \mapsto f(hg)$, so we will smear out our operators following the philosophy of functional analysis. Then we will have

$$f(h) \mapsto \int_G c(g)f(hg) dg,$$

where c is an arbitrary function. The key point will be to show that this operator is **compact** and self-adjoint if $c(g^{-1}) = \overline{c(g)}$. Here a compact operator A is compact if the closure of the image of the unit ball is compact. Here are some basic properties:

1. Compact operators form a two-sided closed ideal in all bounded operators.
2. A is compact if and only if there exist a sequence A_n of finite-rank operators such that $\|A - A_n\| \rightarrow 0$. Each A_n is given by choosing finitely many dimensions and projecting there.
3. If A is compact and self-adjoint, then $\mathcal{H} = \bigoplus \mathcal{H}_{\lambda_i}$, where λ_i are real eigenvalues, $\lambda_i \rightarrow 0$ as $i \rightarrow \infty$, and $\dim \mathcal{H}_{\lambda_i} < \infty$.
4. If A is compact and general, then

$$A = \sum \lambda_i (\psi_n, -) \varphi_n,$$

where $(\psi_n, \psi_m) = (\varphi_n, \varphi_m) = \delta_{nm}$.

The main point is that integral operators are typically compact. Modulo this, we have proved Peter-Weyl.

Now we need to show that $\left(\widehat{\bigoplus} V^* \boxtimes V\right)^\perp = 0$. To do this, we will use spectral decomposition for operators that come from the right action of G . These are compact and self-adjoint operators. Recall that for Hilbert space $\mathcal{H}$ and compact and self-adjoint operator K , then we can write

$$\mathcal{H} = \bigoplus_{\lambda} \mathcal{H}^\lambda,$$

¹Andrei had no idea how this name came to be, so we looked at Wikipedia in real time and found that he was a Dutch physicist. We still have no idea why the name was given.

where $\lambda \neq 0$ and $\dim \mathcal{H}^\lambda < \infty$. Here, we will write

$$[K(f)](g) = \int_G f(gh)c(h) dh$$

where $c(h)$ is a kernel function. Then recall that if $[M(i, j)]$ is a matrix, then we have

$$M \cdot v(i) = \sum_j M(i, j)v(j).$$

Given two spaces X, Y , then for some kernel function $K \in L^2(X \times Y)$, we can define an integral operator

$$[K(f)](x) = \int_Y dy K(x, y)f(y).$$

Now we have the inequality

$$\begin{aligned} \|K(f)\|_{L^2(X)}^2 &= \int_X |K(f)|^2 dx \\ &\leq \int_X dx \left[\left(\int_Y dy |f(y)|^2 \right) \left(\int_Y dy |K(x, y)|^2 \right) \right] \\ &\leq \|f\|_{L^2(Y)}^2 \|K\|_{L^2(X \times Y)}^2, \end{aligned}$$

and thus K defines a bounded operator on X .

Remark 3.2.7. Recall that if V is a finite-dimensional vector space, we have

$$V^* \otimes V \xrightarrow{\cong} \text{End}(V).$$

More generally, for finite dimensional vector spaces, we have

$$V_1^* \otimes V_2 \xrightarrow{\cong} \text{Hom}(V_1, V_2).$$

If $\mathcal{H}_1, \mathcal{H}_2$ are Hilbert spaces, then we have a map

$$\mathcal{H}_1^* \hat{\otimes} \mathcal{H}_2 \rightarrow B(\mathcal{H}_1, \mathcal{H}_2), \quad v_1 \otimes v_2 \mapsto (-, v_1)v_2.$$

This is not surjective and goes into an ideal of Hilbert-Schmidt operators, which are those that have finite L^2 norm under the norm $\sum |M_{ij}|^2$. Then we can write

$$K(x, y) = \sum_{ij} \varphi_i(x)\psi_j(y)$$

and see that $\|K\|^2 = \sum |K_{ij}|^2$.

Remark 3.2.8. Consider the two kernels $X_1 \xrightarrow{K_{21}(x_2, x_1)} X_2 \xrightarrow{K_{32}(x_3, x_2)} X_3$. Then the composition has kernel

$$\int_{X_2} dx_2 K_{32}(x_3, x_2)K_{21}(x_2, x_1).$$

This is analogous to matrix multiplication.

Remark 3.2.9. Consider a measure space (X, dx) . Then the assignment

$$(X, dx) \longrightarrow \text{Functions}$$

is a functor. We have the pullback as usually defined, but we also have pushforwards defined by integration with respect to dx . Here, analytic issues with this integration process are ignored. This generalizes to the structure

$$\begin{array}{ccc} & F(X_1 \times X_2) & \\ p_1^* \nearrow & & \searrow (p_2)_* \\ F(X_1) & & F(X_2) \end{array}$$

where we have $Kf = (p_2)_*(p_1^*(f) \otimes K)$. In the context of sheaves on algebraic varieties, this gives a Fourier-Mukai transform.

We now return to our compact operator K . We know that $\sum K_{ij}\psi_i(x)\varphi_j(y)$ converges. Then we can write

$$K \cdot f = \sum_{i,j} K_{ij}\psi_i(x) \int_Y f(y)\varphi_j(y) dy.$$

Thus K is a limit in the operator norm of operators of finite rank and thus K is compact. The particular operator we want is

$$f \mapsto \int_G f(gh)c(h) dh = \int_G f(h)c(g^{-1}h) dh.$$

This is self-adjoint if $c(g^{-1}) = \overline{c(g)}$. However, we don't need to worry about this because if K is compact and commutes with left translation then K^*K is compact and self-adjoint. Now we use the fact that because the sum of the nonzero eigenspaces of operators like K^*K are dense in $L^2(G)$ and thus the image of operators of this form are dense. Thus if $c(g) \rightarrow \delta(e)$, then

$$\int f(gh)c(h) dh \rightarrow f(g),$$

where δ is the Dirac delta distribution. Thus any function is in the closure of the image. Here, convergence here is convergence in the weak sense in the space of distributions $C^\infty(G)^\vee$. This concludes the proof of Peter-Weyl.

Remark 3.2.10. Recall that we have the inclusion

$$\bigoplus_V V^* \boxtimes V \subset \widehat{\bigoplus_V V^* \boxtimes V} = L^2(G).$$

Then note that the finite direct sum is an **algebra**. Here, we simply note that matrix elements of V_1 times matrix elements of V_2 are matrix elements of $V_1 \otimes V_2$. This is finitely generated (by matrix elements of any faithful representation and its dual). Therefore, the space of functions has a finitely generated dense subset that is an algebra.

In our Lie group $G \subset G_{\mathbb{C}}$ with Lie algebra $\text{Lie}(G) \otimes \mathbb{C}$, recall that $G_{\mathbb{C}}$ is a complex Lie group and is thus analytic. In fact, we will see that $G_{\mathbb{C}}$ is an affine algebraic group, and thus $\bigoplus_V V^* \boxtimes V$ is the algebra of functions on $G_{\mathbb{C}}$. This tells us that every compact Lie group is the real form of a complex algebraic group.

3.3 Representation Theory of Unitary Groups

We have been discussing the Peter-Weyl theorem, and now we will apply this to study the representation theory of compact Lie groups, and in particular, the most important such group $U(n)$. Recalling that

$$L^2(G) = \widehat{\bigoplus_{V \text{ irrep}} \text{End}(V)},$$

there is a distinguished element of each factor: the identity 1_V . This is invariant under $G \subset G \times G$ and corresponds to the character $\text{tr}_V g \in C^\infty(G)$ of V . Recalling that the metric on $\text{End}(V)$ was

$$(A_1, A_2) = \frac{1}{\dim V} \text{tr } A_1 A_2^\dagger,$$

we see that characters of irreducible representations are orthonormal. Taking invariants, we now see

$$L^2(G/\text{conjugation}) = \widehat{\bigoplus_{\text{irreps } V} \mathbb{C} \cdot \chi_V(g)}.$$

For $U(n)$, we will describe this space of functions and the lattice $\bigoplus_V \mathbb{Z} \cdot \chi_V(g)$ explicitly. Then we will find an orthonormal basis of this lattice. We will use the basic fact that $O(n, \mathbb{Z})$ is generated by S_n and ± 1 .

Now recall that by the spectral theorem, all unitary matrices can be diagonalized. Thus we have

$$U(n)/\text{conj} = \left\{ \begin{pmatrix} t_1 & & \\ & \dots & \\ & & t_n \end{pmatrix} \right\} / S_n.$$

This set of diagonal matrices is usually denoted by T and is isomorphic to $U(1)^n$. This is a maximal torus. In general this S_n is replaced by the Weyl group. To draw an explicit picture in the case of $SU(2)$, we have

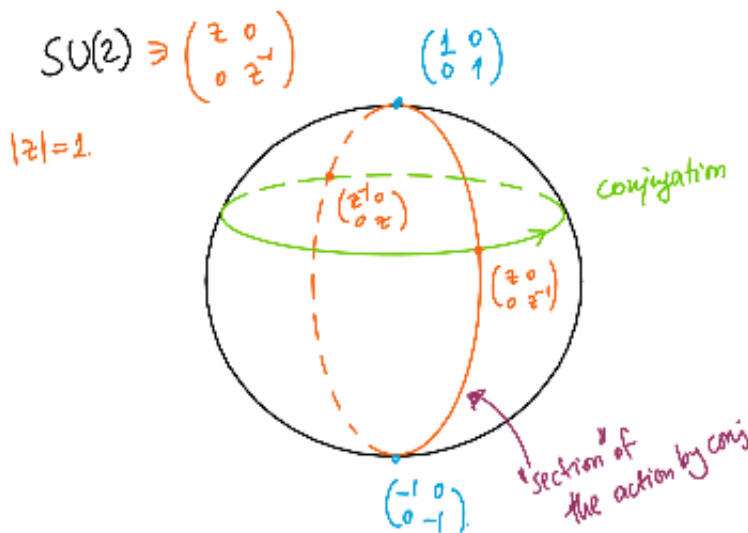


Figure 3.1: Conjugation action of $SU(2)$ on itself

and then we see that

$$N(T) = \left\{ g \mid gTg^{-1} \subset T \right\} = \left\{ (*, *), \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \right\}$$

and then the Weyl group is

$$N(T)/T = \left\{ \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}.$$

For $U(n)$, $N(T)$ is the monomial matrices (or rook placements) and $N(T)/T$ is the set of permutations, or S_n .

Returning to $SU(2)$, we then see that all orbits are parameterized by the eigenvalue $z = e^{i\varphi}$, and each orbit is an S^2 with radius $\sin \varphi$. Thus we have

$$L^2(U(n))^\Delta = L^2(T, \text{interesting measure})^{S_n}.$$

This measure is some constant multiple of $\sin^2 \varphi \, d\varphi$, which is the same as

$$\frac{1}{|W|} (1 - z^2)(1 - \bar{z}^2) \frac{dz}{2\pi iz}.$$

Among all functions on the torus, we may consider the functions given by the coordinates, so we then have a lattice $\mathbb{Z}[t_i^\pm]^{S_n}$. Then we note that this contains $\chi_V(g)$ for all V because as a representation of T , we can split V into 1-dimensional representations $t^\mu = \prod t_i^{\mu_i}$, where $\mu_i \in \mathbb{Z}$. These μ are called the *weights* and the number of times a weight μ appears is the *multiplicity*. Now all unitary matrices are diagonalizable, and thus it suffices to compute characters on T . Therefore we have

$$\chi_V(t) = \sum_{\mu} \text{mult}_V(\mu) \cdot t^\mu.$$

We see each multiplicity is a W -invariant and nonnegative. For $SU(3)$, we have $\mu \in \mathbb{Z}^3/\mathbb{Z}(1, 1, 1)$. In general, the set $\mu_1 \geq \mu_2 \geq \dots \geq \mu_n$ is a fundamental domain for S_n .

Theorem 3.3.1. *For any dominant μ , there exists V^μ such that*

$$\chi_{V^\mu}(t) = t^\mu + \text{lower order terms}$$

with respect to a certain ordering on monomials.

Example 3.3.2. For $SU(2)$, we have

$$\chi_{S^m \mathbb{C}^2} \left(\begin{pmatrix} z & \\ & z^{-1} \end{pmatrix} \right) = z^m + z^{m-2} + \dots + z^{-m}.$$

Then if we consider the adjoint action of $\begin{pmatrix} t_1 & & \\ & \ddots & \\ & & t_n \end{pmatrix}$ on $\mathfrak{gl}(n, \mathbb{C})$, and the weights for this action are called *roots*. Differentiating this, we can consider roots on $\text{Lie}(T)$ and the pairing

$$\left\langle \begin{pmatrix} \xi_1 & & \\ & \dots & \\ & & \xi_n \end{pmatrix}, \delta_i - \delta_j \right\rangle = \xi_i - \xi_j.$$

Then the weight lattice contains the root lattice, which contains the cone spanned by positive roots ($i < j$).

Theorem 3.3.3. *For a dominant weight $\mu = (\mu_1 \geq \dots \geq \mu_n) \in \mathbb{Z}^n/S_n$, there exists V^μ such that*

$$\chi_{V^\mu}(t) = t^\mu + \sum_{\eta \text{ negative}} *t^{\mu+\eta}.$$

Here, we write $\mu > \nu$ if $\mu - \nu$ is in the cone spanned by the positive roots $(0, \dots, 0, 1, 0, \dots, 0, -1, 0, \dots, 0)$. In particular, this means that $t^\mu > t^\nu$ if and only if $t^{\mu-\nu} < 1$, which is the same thing as $t^{\mu-\nu} \rightarrow 0$.

Proof. We will first consider a certain infinite-dimensional representation of $\mathfrak{gl}(n) \supset \mathfrak{u}(n) = \text{Lie}(\mathcal{U}(n))$. It has to have a vector v_μ such that $t \cdot v_\mu = t^\mu v_\mu$. Then for $i < j$, we know that $E_{ii}v_\mu = \mu_i v_\mu$ and $E_{ij}v_\mu = 0$ because this is a vector of weight strictly greater than μ . Now we define $M(\mu)$ to be the free module generated by these relations.

Note that a “free module” for $\mathfrak{gl}(n)$ behaves like the universal enveloping algebra² $\mathcal{U}\mathfrak{gl}(n)$, which is simply $\mathbb{C}\langle E_{ij} \rangle / (xy - yx = [x, y])$ by the Poincaré-Birkhoff-Witt theorem. As a linear space, these are monomials in E_{ij} ordered arbitrarily. In this point of view, we now have

$$M(\mu) = \mathcal{U}\mathfrak{gl}(n) \otimes_{\mathcal{U}(\mathfrak{b})} \mathbb{C} \cdot v_\mu,$$

where $\mathfrak{b}$ is the Lie algebra of upper-triangular matrices and $\mathbb{C}v_\mu$ is a one-dimensional representation of $\mathfrak{b}$ given by $E_{ii}v_\mu = \mu_i v_\mu$ and $E_{ij}v_\mu = 0$ for $i < j$. As a linear space, $M(\mu)$ is spanned by things like $E_{53}E_{74} \cdots v_\mu$, which are arbitrary products of E_{ij} for $i > j$. Then if $i \leq j$, we can write

$$E_{ij} \cdot E_{53}E_{74}E_{21}v_\mu = E_{53}E_{74}E_{21}E_{ij}v_\mu$$

using the commutation relations, so for this mildly noncommutative algebra, we have a Gröbner basis.

Thus the Verma module $M(\mu)$ has the form $t^\mu + \sum_{\nu < \mu} m(\mu - \nu)t^\nu$, where $m(\mu - \nu)$ is the number of ways to write $\mu - \nu$ as a sum of positive roots. As a generating function, this becomes

$$\begin{aligned} \chi_{M(\mu)}(t) &= \frac{t^\mu}{\prod_{\alpha > 0} (1 - t^{-\alpha})} \\ &= \frac{t^\mu}{\prod_{i > j} (1 - t_i/t_j)}. \end{aligned}$$

Now consider the following remarks:

1. We can define $M(\mu)$ for any $\mu \in \mathbb{C}^n$ by $E_{ii}v_\mu = \mu_i v_\mu$. For generic μ , this is irreducible.
2. For positive dominant weights μ , this is always reducible. If $M' \subset M(\mu)$ is a submodule, then the weights of M' contain μ if and only if $M' = M$. Thus $M(\mu)$ contains a maximal proper submodule M' , and we call $L(\mu) = M(\mu)/M'$ to be the irreducible module with *highest weight* μ . Then we can write $\chi_{L(\mu)}(t) = t^\mu + \text{lower order terms}$.

We now need to show that $\dim L(\mu) < \infty$ by relating it to the group. The most direct way is through matrix elements. Write $\varphi(g)$ to be the coefficient of v_μ in $g \cdot v_\mu$. For diagonal matrices $t \in T$, we have $\varphi(t) = t^\mu$. For the upper-triangular matrices, we have $\varphi(gu_+) = \varphi(g)$. Then for any lower-triangular u_- , we have $\varphi(u_-) = 0$. Finally, because generic matrices have a Gauss decomposition, the set $U_- T U_+$ is dense in $GL(n)$. Thus the function is determined uniquely by the values we already have.

²This is also the algebra of right-invariant differential operators on $GL(n)$.

Now because the minor of the $i \times i$ submatrix given by the first i rows and first i columns is invariant under U_i and U_+ , we see that

$$\varphi = \Delta_1^{\mu_1 - \mu_2} \Delta_2^{\mu_2 - \mu_3} \dots \in (V^\mu)^* \boxtimes V^\mu \subset L^2(U(n)).$$

Considering the span under the right regular representation, we obtain V^μ . This span is finite dimensional because it is contained in the space of polynomials. $\square$

Corollary 3.3.4. 1. *We have the identity*

$$\bigoplus_V \mathbb{Z}_{\chi_V}(t) = \mathbb{Z}[t_1^\pm, \dots, t_n^\pm]^{S_n}.$$

2. *We can compute $\chi_V^\mu(t)$ by Gram-Schmidt because only finitely many weights are smaller. This will follow from orthogonality of $\chi_{V^\mu}, \chi_{V^\nu}$ whenever $\nu < \mu$.*
3. *There is a formula for this inner product, and orthogonality can be done explicitly in one step.*

Now we want to prove the *Weyl integration formula*. Let f be a conjugation-invariant function on $G = U(n)$. Then we have

$$\int_G f(g) dg = \frac{1}{n!} \int_T f(t) \prod_{i < j} |t_i - t_j|^2 dt,$$

where $dt = \prod_k \frac{dt_k}{2\pi i t_k}$ is the Haar measure on T .

Remark 3.3.5. For $SU(2)$, recall that for $t = e^{i\varphi}$, we have $\sin^2 \varphi \propto |t - t^{-1}|^2$.

Consider the map $G/T \times T \rightarrow G$ given by $(g, t) \mapsto gtg^{-1}$. This is an $n!$ -to-1 map, and so we can write

$$\int_G f(g) dg = \frac{1}{n!} \int_{G/T \times T} f(t) dt dg/t \cdot \mathcal{J}(t),$$

where $\mathcal{J}$ is some Jacobian. Because the Haar measure is invariant under conjugation, this Jacobian is independent of g . Now computing at the point $g = 1$, we need to compute

$$\begin{aligned} (1 + \delta x)(t + t\delta t)(1 + \delta x)^{-1} &= 1 + (t\delta t + \delta x t - t\delta x) + \dots \\ &= 1 + t(\delta t + t^{-1}\delta x t - \delta x) + \dots \end{aligned}$$

The linear term is $\delta t + (\text{Ad}(t^{-1}) - 1)\delta x$, so we need to compute $\det(\text{Ad}(t^{-1}) - 1)$, but this operator has eigenvalues $t^{-\alpha} - 1$, where α is a root. Therefore, we have

$$\mathcal{J} = \prod_{\alpha} (t^{-\alpha} - 1) = \prod_{\alpha > 0} (t^{\alpha} - 1)(t^{-\alpha} - 1) = \prod_{\alpha > 0} |t^{\alpha} - 1|^2.$$

Now recall that $T \hookrightarrow G \rightarrow G/T$ is a locally trivial fibration. Suppose we have a section s . Then the map $m: G/T \times T \rightarrow G$ given by $(g, t) \mapsto gt$ is generically one-to-one, so we can pull back the Haar measure. This gives an invariant measure on $G/T \times T$. Because the Haar measure is bi-invariant, the measure is invariant on both T and G/T . On the other hand, the conjugation map c is generically $n!$ -to-1, and thus we have

$$c^*(\text{Haar measure}) = \frac{1}{n!} (\text{invariant measure on } G/T) \times (\text{interesting measure on } T).$$

Now if dg is the Lebesgue measure on $\mathfrak{g}$, then we have

$$\int_G f(g) dg = \frac{1}{n!} \text{vol}(G/T) \int_T f(t) \prod_j \frac{dt_j}{it_j}.$$

Normalizing this to have volume 1 using $\text{vol}(G) = \text{vol}(G/T) \cdot \text{vol}(T)$, we see that

$$\int_G f(g) dg = \frac{1}{n!} \int_T f(t) \prod_j \frac{dt_j}{2\pi i t_j}.$$

This proves the Weyl integration formula. To check this, we want to see that

$$1 = \int_G dg = \frac{1}{n!} \int_T \prod_{\alpha > 0} (t^\alpha - 1)(t^{-\alpha} - 1) \prod_j \frac{dt_j}{2\pi i t_j}.$$

But then the integrand can be written as

$$\begin{aligned} \prod_{\alpha} (t^\alpha - 1)(t^{-\alpha} - 1) &= \prod_{i < j} (t_i - t_j)(t_i^{-1} - t_j^{-1}) \\ &= \left(\sum_{w \in S_n} (-1)^w w \cdot t_1^{n-1} t_2^{n-2} \cdot 1 \right) \overline{\left(\sum_{w \in S_n} (-1)^w w \cdot t_1^{n-1} t_2^{n-2} \cdot 1 \right)} \\ &= \frac{1}{n!} n! \\ &= 1 \end{aligned}$$

by the orthogonality relations for monomials on the torus. Now consider the map

$$\mathbb{Z}[t_1^\pm, \dots, t_n^\pm]^{S_n} \xrightarrow{\prod_{i < j} (t_i - t_j)} \{\text{antisymmetric polynomials}\}.$$

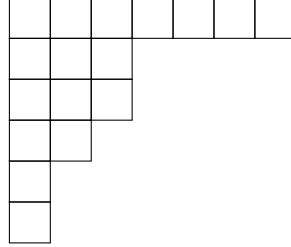
This is in fact an isomorphism over $\mathbb{Z}$. This is because we can divide any antisymmetric polynomial by the difference between any two variables. Writing the Vandermonde determinant as Δ the Weyl inner product with density $\Delta \bar{\Delta} \frac{1}{n!}$ maps to $\frac{1}{n!}$ the usual inner product on $L^2(T)$. Therefore we can write

$$s_\lambda(t_1, \dots, t_n) = \frac{\sum_w (-1)^w w \cdot x_1^{\lambda_1 + n - 1} x_2^{\lambda_2 + n - 2} \dots x_n^{\lambda_n}}{\prod_{i < j} (x_i - x_j)}$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. In the limit where $x_1 \gg x_2 \gg x_3 \gg \dots$, we then see that $s_\lambda = x_1^{\lambda_1} x_2^{\lambda_2} x_3^{\lambda_3} \dots + (\text{lower terms})$ and thus the s_λ form an orthonormal basis of $\mathbb{Z}[t_1^\pm, \dots, t_n^\pm]^{S_n}$. This means that s_λ is in fact an irreducible character. In fact we have given Weyl's proof of

Theorem 3.3.6 (Schur). *The s_λ that we computed above are the characters of irreducible finite-dimensional representations of $U(n), GL(n)$. These are known as Schur functions and were studied by mathematicians like Jacobi and Cauchy.*

In fact, Weyl computed the characters of all compact connected groups in this way. Here, we note that the index $\lambda = (7, 3, 3, 2, 1, 1)$ corresponds to the diagram



which is called its *Young diagram*. Then for example, the function $s_{(1,1,1,1,0,0,\dots)}$ is the character of the irreducible representation with highest weight $(1, 1, 1, 1, 0, 0, \dots)$, i.e. $t_1 t_2 t_3 t_4$ and then $\bigwedge^4 V$ has a basis $e_1, e_2, \dots$, with e_1 corresponding to the largest eigenvalue, e_2 the second largest, etc. In fact, this wedge power is actually irreducible and therefore we have for example

$$s_{\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}}(t) = \sum_{i_1 < i_2 < i_3 < i_4 \leq n} t_{i_1} t_{i_2} t_{i_3} t_{i_4}$$

is the elementary symmetric function $e_4(t)$. Similarly, we can compute

$$s_{\begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \end{array}} = t^4 + \text{lower} = \chi_{S^4 V} = \sum_{i_1 \leq \dots \leq i_4 \leq n} t_{i_1} t_{i_2} t_{i_3} t_{i_4}$$

is the complete homogeneous symmetric function $h_4(t)$. Now we can consider the projective limit and consider partitions as

$$s_\lambda \in \varprojlim \mathbb{Z}[t_1, \dots, t_n]^{S(n)}$$

and this projective limit is the *algebra of symmetric functions* and is taken in the category of graded rings. For more general partitions, we have

$$s_{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \end{array}} = \sum_{\substack{\text{standard Young tableaux} \\ \text{of shape } \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & & \\ \hline \end{array}}} \prod t_{i_k}.$$

Then we have the remarkable *Weyl dimension formula*, which is

$$\dim(\text{irrep of } U(n)) = s_\lambda(1, \dots, 1).$$

This evaluation is computed by evaluating $s_\lambda(1, q, \dots, q^{n-1})$ and then taking $q \rightarrow 1$. We can compute this by

$$s_\lambda = \frac{\det(x_i^{\lambda_j + n - j})}{\det(x_i^{n - j})}$$

and then by noting that

$$\det(q^{(i-1)(\lambda_j + n - j)}) = \det((q^{\lambda_j + n - j})^{(i-1)}),$$

and thus we have

$$s_\lambda(1, q, \dots, q^{n-1}) = \frac{\prod_{i < j} (q^{\lambda_i + n - i} - q^{\lambda_j + n - j})}{\prod_{i < j} (q^{n - i} - q^{n - j})}.$$

Then in the limit as $q \rightarrow 1$, we obtain the limit to be

$$\prod_{i < j} \frac{\lambda_i - \lambda_j + j - i}{j - i} = \dim V_\lambda$$

To rewrite all of this, we have

$$s_\lambda(1, q, \dots, q^{n-1}) = \sum_{\text{standard tableaux}} * = q^{\sum (i-1)\lambda_i} (1 + \dots).$$

Then this can actually be computed in terms of the hook lengths, where the hook length of a box is the number of rows below and columns to the right of the box. In addition, we write $c(\square)$ to be the difference of column number and row number. Thus we have

$$s_\lambda = q^{\sum (i-1)\lambda_i} \prod_{\square} \frac{1 - q^{n+c(\square)}}{1 - q^{h(\square)}}.$$

For fun, we can compute the number of three-dimensional partitions with a given two-dimensional base. These look something like

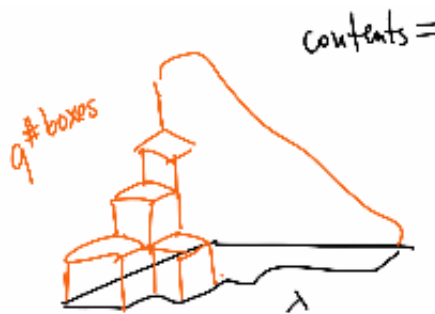


Figure 3.2: A 3-dimensional partition

For a partition $\lambda = (\lambda_1 \geq \lambda_2 \geq \dots \geq 0)$, we will write $|\lambda| = \sum \lambda_i$ and $\ell(\lambda)$ for the number of nonzero entries of λ . Then recall that irreducible representations of $U(n)$ and $GL(n)$ correspond to sequence $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$, where $\lambda_i \in \mathbb{Z}$. Now if we consider only polynomial representations (that involve no $\frac{1}{\det}$), then we have the additional condition that $\lambda_n \geq 0$. To see this, note that if we consider the function

$$\prod_{i=1}^n \Delta_i^{\lambda_i - \lambda_{i+1}} = \Delta_1^{\lambda_1 - \lambda_2} \dots \Delta_{n-1}^{\lambda_{n-1} - \lambda_n} \Delta_n^{\lambda_n},$$

where Δ_i is the i -th minor, then this is a polynomial if and only if the last weight is nonnegative. Therefore, polynomial irreps of $GL(n)$ correspond to partitions λ with length $\ell(\lambda) \leq n$. Now if we set $V = \mathbb{C}^n$, then each polynomial irrep occurs in some $V^{\otimes k}$. This has an action of $S(k)$ and then has the Schur-Weyl decomposition, where we have

$$V^{\otimes k} = \bigoplus_{\substack{|\lambda|=k \\ \ell(\lambda) \leq n}} V^\lambda \boxtimes M^\lambda,$$

where V^λ is the irrep of $GL(n)$ corresponding to λ and M^λ is the irrep of S_k corresponding to λ . For the symmetric group, recall that irreps are in bijection with conjugacy classes for all finite groups and for the symmetric group, conjugacy classes are the same as cycle types, which are the same thing as partitions. If we write $p(k)$ for the number of partitions of k , then there is a formula due to Euler³

$$\sum_{k \geq 0} p(k)q^k = \sum_{\lambda} q^{|\lambda|} = \prod_{n \geq 1} \frac{1}{(1 - q^n)}$$

using the basic theory of generating functions. This series is very close to a modular form, and in fact the series

$$q \prod_{n \geq 1} (1 - q^n)^{24}$$

is the first cusp form. Now if we write $q = e^{2\pi i \tau}$ for $\tau \in \mathcal{H}$, then the series $\sum q^n p(n)$ converges for $|q| < 1$, and if we take the $q \rightarrow 1$ behavior we learn about the asymptotic behavior of the partition function. We then obtain the approximation (due to Hardy and Ramanujan)

$$p(n) \sim \frac{1}{2\pi\sqrt{2}} \sum_{k=1}^{\infty} A_k(n) \sqrt{k} \frac{d}{dn} \left(\frac{1}{\sqrt{n - \frac{1}{24}}} \exp \left[\frac{\pi}{k} \sqrt{\frac{2}{3} \left(n - \frac{1}{24} \right)} \right] \right),$$

where

$$A_k(n) = \sum_{\substack{0 \leq m < k \\ (m,k)=1}} e^{\pi i (s(m,k) - 2nm/k)}.$$

Then for three-dimensional partitions, there is the formula (due to McMahon) that

$$\sum_{\pi \text{ 3D partition}} q^{|\pi|} = \frac{1}{\prod_{n \geq 1} (1 - q^n)^n}.$$

To prove this formula, we first count the number of partitions that sit in a given box. In the two dimensional case, if we have a $k \times m$ rectangle, then the function

$$\sum_{\lambda \subset \square} q^\lambda = \# \text{Gr}(m+k, k)(\mathbb{F}_q).$$

This is the q -analog of $\binom{m+k}{k}$. Now in three dimensions, for a box of dimensions $k \times m \times n$, we obtain the formula

$$\prod_{i=1}^k \prod_{j=1}^m \frac{1 - q^{i+j+n-1}}{1 - q^{i+j-1}}.$$

In particular, when $n = k = m = \infty$, we obtain the formula

$$\prod_{i,j} \frac{1}{1 - q^{i+j-1}} = \prod_n \frac{1}{(1 - q^n)^n}.$$

In fact, we can generalize this to skew 3-dimensional partitions that look like

³Euler knew nothing about representation theory of algebraic geometry, so Andrei has no idea why he, Hardy, or Ramanujan cared about partitions.

$H \not\subset \text{Im}(\tilde{H})$ near 1, then there exist $p_n \in \mathfrak{p}$ where $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{p}$ such that $\exp p_n \in H, p_n \rightarrow 0$. Now take a convergent sequence among the $\frac{p_n}{\|p_n\|} \rightarrow \xi$. We have

$$\begin{aligned} e^{t\xi} &= \lim_{n \rightarrow \infty} e^{t \frac{p_n}{\|p_n\|}} \\ &= \lim_{n \rightarrow \infty} e^{\left[\frac{t}{\|p_n\|} \right] p_n + \left\{ \frac{t}{\|p_n\|} \right\} p_n} \end{aligned}$$

and then note that the integer part is in H and the fractional part converges to 0. $\square$

Proposition 3.4.2. *Maximal tori are in one-to-one correspondence with maximal abelian subalgebras.*

We will give two different proofs of this fact.

First Proof. Let $\mathfrak{t} \subseteq \mathfrak{g}$ be maximal commutative of dimension r . Then we have a morphism $\mathbb{R}^r \rightarrow G$ of Lie groups, so we need to prove the image is closed. If the image is not closed, then its image is a closed connected abelian subgroup of dimension strictly larger than r . Thus its Lie algebra strictly contains $\mathfrak{t}$ and is abelian, and this contradicts maximality of $\mathfrak{t}$. $\square$

Second Proof. Let G be a Lie group (or an affine algebraic group). Let $\mathfrak{h} \subset \mathfrak{g} = \text{Lie}(G)$ be a subalgebra. Then let $\bar{\mathfrak{h}}$ be the Lie algebra of the minimal closed subgroup $\bar{H} \subset G$ such that $\text{Lie}(\bar{H}) \supset \mathfrak{h}$. Then we will show that

$$[\bar{\mathfrak{h}}, \bar{\mathfrak{h}}] = [\mathfrak{h}, \mathfrak{h}].$$

To prove this, consider

$$H_1 = \left\{ h \in G \mid \text{Ad}(h)\mathfrak{h} \subset \mathfrak{h}, \text{Ad}(h) \Big|_{\mathfrak{h}/[\mathfrak{h}, \mathfrak{h}]} = \text{id} \right\}_0.$$

This is a closed Lie/algebraic subgroup. Next, we see that

$$\text{Lie}H_1 = \left\{ \xi \in \mathfrak{g} \mid [\xi, \mathfrak{h}] \subseteq \mathfrak{h}, \text{ad}(\xi) \Big|_{\mathfrak{h}/[\mathfrak{h}, \mathfrak{h}]} = \text{id} \right\}.$$

This implies that $\bar{H} \subset H_1$. Now we do the same for

$$\text{Lie}(H_2) = \{ \xi \in \mathfrak{g} \mid \text{ad}(\xi)\mathfrak{h}_1 \subset \mathfrak{p}\mathfrak{h}, \mathfrak{h} \}.$$

This contains $\mathfrak{h}$ because $[\mathfrak{h}_1, \mathfrak{h}] \subset [\mathfrak{h}, \mathfrak{h}]$ and therefore $\bar{H} \subset H_2$ and thus $[\bar{\mathfrak{h}}, \bar{\mathfrak{h}}] \subset [\bar{\mathfrak{h}}, \mathfrak{h}_1] \subset [\mathfrak{h}, \mathfrak{h}]$. $\square$

Now consider the following observations:

1. $T = \mathbb{R}^n/\mathbb{Z}^n$ has no continuous automorphisms because $\text{GL}(n, \mathbb{Z})$ is discrete. Then we have an exact sequence

$$1 \rightarrow C_G(T) \rightarrow N_G(T) \rightarrow \text{Aut}(T)$$

is exact. We know that $C_G(T)_0 = T$ by maximality, and in fact we will see that $C_G(T) = T$. In addition, we know that $W = N(T)/T$ is discrete and compact, and hence finite.

2. Because T has only countably many Lie subgroups, we have

$$T = \left\{ \begin{array}{ll} \overline{\bigcup_n t^n} & t \in T \text{ random} \\ \exp(\xi) & \xi \in \text{Lie}(T) \text{ random} \end{array} \right\}.$$

Theorem 3.4.3. *Let $T \subset G$ be a maximal torus. Then*

1. *Any element $g \in G$ is conjugate to an element of T .*
2. *Any $\xi \in \text{Lie}(G)$ is conjugate to an element of $\text{Lie}(T)$.*

Corollary 3.4.4. *All maximal tori are conjugate.*

Proof. Let T' be a different maximal torus. Then choose $t \in T'$ such that $\overline{\bigcup_n t^n} = T'$. Then by the theorem, there exists g such that $gtg^{-1} \in T$. Thus $gt^n g^{-1} \in T$ and thus $gT'g^{-1} \subset T$ and maximality implies that this is actually an equality. $\square$

There is a general principle for proving theorems of this kind. If $H \subset G$, then $g^{-1}tg \in G$ is the same as $tg \in gH$ and thus $tgH = gH$ as sets. This implies that t has a fixed point gH on the manifold G/H . More generally, a subgroup H' being conjugate to a subgroup of H is equivalent to H' fixing a point gH of the manifold G/H .

Now we consider the *flag manifold (or flag variety)* G/T for G . For $U(n)$, we see that T is the centralizer of a generic diagonal matrix. We can describe G/T as the generic class in either $U(n)$ or $u(n)$. This is alternatively the moduli space of orthonormal frames in $\mathbb{C}^n$. Equivalently, this is the moduli space of flags $V_1 \subset V_2 \subset \cdots \subset V_n = \mathbb{C}^n$ (hence the name *flag variety*).

If I draw the flag red, will it be too communist? I don't want to offend anyone.

A. Okounkov

It is easy to see that $GL(n, \mathbb{C})$ acts transitively on the flag variety and that the stabilizer of a given flag is the group B of upper-triangular matrices (known as the Borel). This implies that the flag variety is a compact complex manifold. For example, we see that $U(2)/T = GL(2, \mathbb{C})/B = \mathbb{CP}^1$. A key property is that the general flag variety G/T has a cell decomposition into $|W|$ -many cells, each of even dimension, called *Schubert cells*. In particular, this means that $\pi_1(G/T) = 0$. Next, if $t: M \rightarrow M$ is homotopic to the identity, then a very special case of the Lefschetz fixed-point formula says that

$$\sum_{t(m)=m} \text{mult}(m) = \chi(M)$$

where the multiplicity is the intersection number $\Gamma(t) \cdot \Delta$ in $M \times M$. Because $\chi(G/T) > 0$, there must exist a fixed point of positive multiplicity. Therefore, because any element of G is homotopic to the identity, it must have $\chi(G/T) > 0$ fixed points on G/T counting multiplicity.

We will prove that any element of a compact connected Lie group G is conjugate to something in a maximal torus. Consider the adjoint action of a fixed maximal torus H (or its Lie algebra $\mathfrak{h}$) on $\mathfrak{g}$. Then the adjoint action of H on $\mathfrak{g} \otimes \mathbb{C}$ splits into a direct sum of eigenspaces with nontrivial eigenvalues the roots. Thus we have

$$\mathfrak{g} \otimes \mathbb{C} = \mathfrak{h} \otimes \mathbb{C} \oplus \bigoplus_{\alpha \neq 0} \mathfrak{g}_\alpha.$$

Now choose $\xi \in \mathfrak{h}$ generic such that $\alpha(\xi) \neq 0$ for any root α . We want to prove

Theorem 3.4.5. *Any $x \in \mathfrak{g}$ is conjugate to an element of $\mathfrak{h}$.*

This is implied by

Proposition 3.4.6. *Any $x \in \mathfrak{g}$ is conjugate to an element that commutes with ξ .*

Proof. Consider the adjoint orbit $O(x) \subset \mathfrak{g}$ of x , which is a compact smooth manifold. Consider a function $f(y) = (\xi, y)$ on $O(x)$ where, $(-, -)$ is the invariant metric on $\mathfrak{g} = \text{Lie}(G)$. Because $O(x)$ is compact, f has a critical point and thus $(\xi, -)$ vanishes on the tangent space to that point y . This is the span of $[\eta, y]$ for $\eta \in \mathfrak{g}$. Thus there exists y such that for all η , we have

$$0 = (\xi, [\eta, y]) = (\xi, [y, \eta]) = ([\xi, y], \eta)$$

and thus $[\xi, y] = 0$. □

We remark that $(-, -)$ gives a G -equivariant isomorphism $\mathfrak{g} \simeq \mathfrak{g}^*$. Even without this form, one can still talk about $\mathfrak{g}^*$, which is a very important object. In fact, the Lie bracket on $\mathfrak{g}$ gives a *Poisson bracket* on $\mathfrak{g}^*$.

Definition 3.4.7. A manifold M is a *Poisson manifold* if there exists a Poisson bracket $\{f, g\}$ is the structure of a Lie algebra on functions $\mathcal{O}(M)$ that is compatible with the product by the Leibniz rule. Here, this means that

$$\{f_1 f_2, g\} = \{f_1, g\} f_2 + f_1 \{f_2, g\}.$$

If the product is commutative, this means that we have a derivation of the product and thus we have a vector field.

Note that $\mathfrak{g}^*$ has coordinates $\mathfrak{g}$. Thus we can extend $\{-, -\}$ from $[_, _]$ by the Leibniz rule, and this turns $\mathfrak{g}^*$ into a Poisson manifold. This is known as the *Lie-Kirillov-Kostant* structure.

For a Poisson manifold M , the span of $\{g, -\} \subset T_m M$ satisfies the integrability criterion. The Jacobi identity tells us that that

$$\{g_1, \{g_2, f\}\} - \{g_2, \{g_1, f\}\} = \{\{g_1, g_2\}, f\}.$$

Then the integral manifold passing through m is called the *symplectic leaf* of m and is a locally closed submanifold of M .

On $\mathfrak{g}^*$, we consider the subspace $\text{span}(\text{ad}(\eta)^*)x$, and thus the integral manifold is actually the G -orbit of x . This orbit is called the *coadjoint orbit* and is equal to the adjoint orbit for compact groups.

Then we have for every symplectic leaf

$$C^\infty(\text{Leaf}) = C^\infty(M)/\text{Poisson ideal}$$

and therefore $T_m^* \text{Leaf} \xrightarrow{\{g, -\}} T_m \text{Leaf}$ is an isomorphism. Thus the map gives us a 2-form on $T_m \text{Leaf}$ which is antisymmetric by the axioms of the Poisson bracket. Now the axioms of the Poisson algebra mean that $\omega \in \wedge^2 T_m^* \text{Leaf}$ is closed, and thus the leaf is a symplectic manifold.

Definition 3.4.8. A *symplectic manifold* is a manifold with a closed nondegenerate 2-form ω .

Theorem 3.4.9 (Kirillov-Kostant). *Coadjoint orbits of a Lie group are symplectic manifolds.*

Now we will discuss connections to Morse theory. The idea is that a Morse function (function with all critical points nondegenerate) on a compact manifold tells us a lot about the topology of the manifold. Here, a Morse function is precisely a function where $\nabla f = 0$ implies $\nabla^2 f$ is nondegenerate. An example of a Morse function is the height function on a torus:

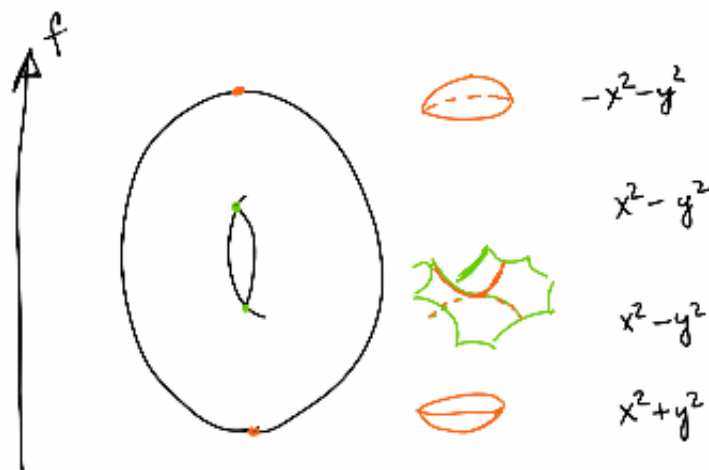


Figure 3.4: Morse function on a torus.

Now we can rebuild the manifold by gluing on handles, which are $D^i \times D^{n-i}$, where n is the dimension of the manifold and the attaching map is on the D^i part. This allows us to reconstruct the torus as follows:

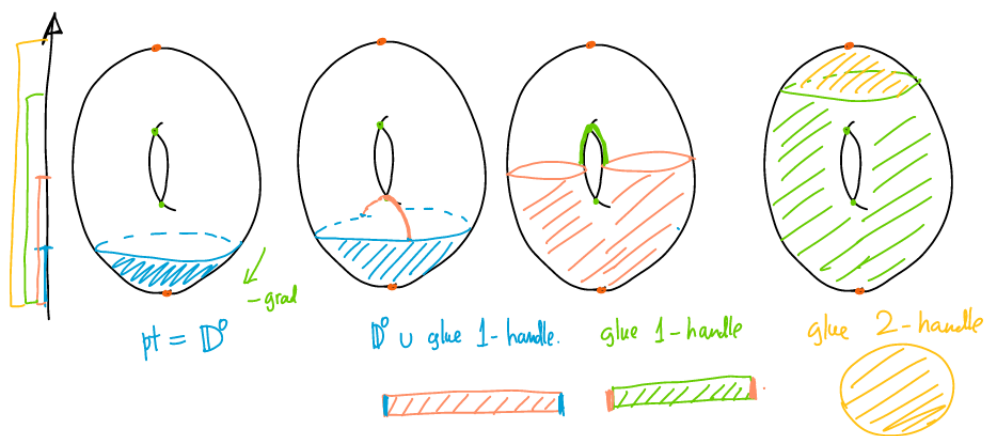


Figure 3.5: Handle decomposition of torus.

Here at each critical point we are gluing a k -handle, where k is the index. From the symplectic point of view, the main property of $(\xi, -)$ is that the index of any critical point is even. This gives us a cell decomposition into cells of even dimension, and therefore $\pi_1(O(x)) = 0$.

Corollary 3.4.10. *The centralizer of C_x of x is connected.*

Proof. Note that $C_x \rightarrow G \rightarrow O(x)$ is a locally trivial fibration. Therefore we have an exact sequence

$$0 \rightarrow \pi_0(C_x) \rightarrow \pi_0(G) \rightarrow \pi_0(O(x)) \rightarrow 1,$$

and thus $\pi_0(C_x) = 0$. □

Note that $\pi_2(O(x)) = H^2(O(x))$ by Hurewicz and that $H^2(O(x))$ is in bijection with critical points of index 2. Because of the cell decomposition into cells of even dimension, we have $H^\bullet(O(x)) = \mathbb{Z}[\text{cells}]$. This also means that $(\xi, -)$ has a unique maximum and minimum.

Note that if x is generic, then $C(x)$ is a maximal torus. This is because for generic $\xi \in H$, we have $C(\xi)_0 = H$. Thus generic adjoint/coadjoint orbits of G are of the form G/H .

Returning to conjugacy in a Lie group, we can either use the Lefschetz formula to conclude that $g \in G$ has a fixed point on G/H or we can write $g = \exp(x)$.

3.5 Poisson Geometry of Lie Groups

We will say some more about Poisson manifolds. Recall that these are manifolds with a Lie algebra structure $\{-, -\}$ on $C^\infty(M)$ that satisfies the Leibniz rule. As a subclass of these, we have symplectic manifolds (M, ω) , where $\omega \in \Omega^2(M)$ is closed and nondegenerate. Then the two maps

$$v \mapsto \omega(v, -) \quad f \mapsto \{f, -\}$$

are inverse to each other. Locally, all symplectic manifolds look the same, and there are always $p_1, \dots, p_n, q_1, \dots, q_n$ such that $\{p_i, q_j\} = \delta_{ij}$. This is a remarkable contrast to the Riemannian case, where Riemannian manifolds have local invariants such as curvature. On the other hand, Poisson manifolds may locally be very complicated. For example, if $\{x_i, x_j\}$ vanishes at $x = 0$, then the next term looks like a Lie algebra, so the classification of Poisson manifolds looks like the classification of Lie algebras. Thus we will focus on symplectic manifolds, which originated in classical mechanics. Here, the q_i are coordinates and p_i are momenta. Then the equations of motion in Hamiltonian form are

$$\frac{d}{dt} f = \{\text{Hamiltonian}, f\}$$

where the Hamiltonian is the total energy of the system. For example, if we consider a harmonic oscillator $q \in \mathbb{R}$, then we have $H = \frac{p^2}{2} + \frac{q^2}{2}$ and thus

$$\frac{d}{dt} q = \{H, q\} = \left\{ \frac{p^2}{2}, q \right\} = p \quad \frac{d}{dt} p = \{H, p\} = \left\{ \frac{q^2}{2}, p \right\} = -q.$$

Therefore the orbits (going around a circle clockwise) of this motion are periodic with period 2π . Now if we multiply H by a constant, this will modify the angular velocity. Now if

$$H = \sum_i \frac{c_i}{2} (p_i^2 + q_i^2) \quad \{p_i, q_j\} = \delta_{ij},$$

then the motion will rotate each q_i, p_i -plane with angular velocity c_i .

In our case, our maximal torus T acts on the (co)adjoint orbit $O(x) \subset \mathfrak{g}$. Now each 1-parameter subgroup $\exp(t\xi)$ acts via the Hamiltonian $(\xi, -)$. Then the fixed points of this action are critical points of the function. We know $y = gxg^{-1}$ is a fixed point if $[\xi, y] = 0$, where $\xi \in \mathfrak{t}$ is also generic. Also, we may assume that $y \in \mathfrak{t}$ and $g \in N(T)/T = W$. Thus fixed points of T are the same as critical points of $(\xi, -)$, which correspond to the Weyl group.

To compute the index of a critical point, we have $T_y O = \mathfrak{g}/\mathfrak{t}$, and so after complexification, we have

$$\mathfrak{g}_\mathbb{C} = \mathfrak{t}_\mathbb{C} + \bigoplus_{\alpha} \mathfrak{g}_{\alpha}.$$

Because the roots come in complex conjugate pairs, we will have 2-dimensional representations that T rotates with angular velocity α . Now the Poisson bracket is

$$\{\eta_1, \eta_2\}_y = ([\eta_1, \eta_2], y)$$

where $(-, -)$ is the G -invariant inner product. By invariance, we know that $([\eta_1, \eta_2], y) = y$ unless $[\eta_1, \eta_2] \in \mathfrak{t}$.

Lemma 3.5.1. *If $\alpha, \beta, \alpha + \beta$ are roots, then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$.*

Proof. We have

$$\text{Ad}(\mathfrak{t})[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = [\mathfrak{t}^\alpha \mathfrak{g}_\alpha, \mathfrak{t}^\beta \mathfrak{g}_\beta] = \mathfrak{t}^{\alpha+\beta} [\mathfrak{g}_\alpha, \mathfrak{g}_\beta].$$

More generally, if V is a G -module, then $\mathfrak{g}_\alpha V_\beta \subset V_{\alpha+\beta}$ by the same argument. $\square$

Further, the form $(-, -)$ is nondegenerate, so its extension to $\mathfrak{g}_\mathbb{C}$ is nondegenerate. Also, by T -invariance, we have $\mathfrak{g}_\alpha \perp \mathfrak{g}_\beta$ unless $\alpha + \beta = 0$, so $\mathfrak{g}_{-\alpha} = \mathfrak{g}_\alpha^*$ with respect to this metric. Moreover, we have

$$[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{C}h_\alpha$$

where $h_\alpha \in \mathfrak{t}$ corresponds to the linear form $\alpha \in \mathfrak{t}^*$ via $(-, -)$. To prove this, note that

$$(y, [e_\alpha, e_{-\alpha}]) = ([y, e_\alpha, e_{-\alpha}]) = \alpha(y)(e_\alpha, e_{-\alpha}).$$

Next, we have $\{e_\alpha, e_{-\alpha}\}_y = (y, [e_\alpha, e_{-\alpha}]) = \alpha(y)(e_\alpha, e_{-\alpha})$. Thus the sign of the Poisson bracket has to do with positivity of $\alpha(y)$.

Because we have $(\mathfrak{g}/\mathfrak{t})_\mathbb{C} = \bigoplus_\alpha \mathfrak{g}_\alpha$, then the Hamiltonian for the action of ξ is

$$H = \sum_{\text{pairs of roots}} \pm \left(\sum_{i=1}^{\dim \mathfrak{g}_\alpha} p_i^2 + q_i^2 \right),$$

the index is even and is equal to $|\{\alpha > 0 \mid \alpha(y) < 0\}|$ assuming that $\dim \mathfrak{g}_\alpha = 1$. For any x , we can define $\alpha > 0$ if $\alpha(x) > 0$. Then for any fixed point $y = wx$ for some $w \in W$, this index is

$$|\{\alpha > 0 \mid \alpha(wx) < 0\}| = |\{\alpha > 0 \mid w \cdot \alpha < 0\}|$$

And is equal to the number of hyperplanes separating x from wx , which is the number of hyperplanes separating the positive cone from $w(\text{positive cone})$. This is also the length $\ell(w)$, which is the usual number of inversions (adjacent transpositions) for $S(n)$. In summary, the index of a critical point is always even and we have the following dictionary:

Table 3.1: Dictionary between Morse theory and the Weyl group.

Geometry	Algebra
Critical points	Weyl group
index	length

Now we have a general statement: If T acts by a Hamiltonian action on any symplectic manifold M , then the index of any component of the critical locus is even. Geometrically, in the context of Morse theory, something like the below cannot happen:

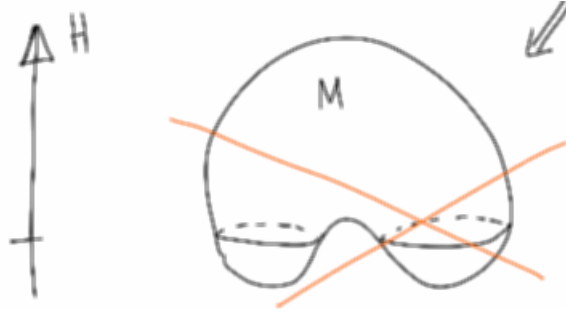
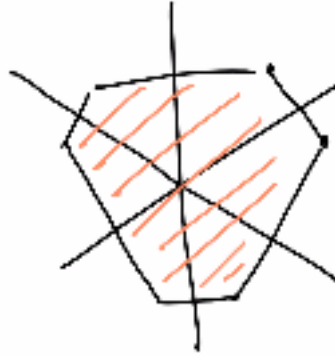


Figure 3.6: An impossible scenario

In particular, every level set of H is connected and M is obtained by always attaching even-dimensional handles (and these have connected boundaries). Now if we have $H_1, \dots, H_r \in \text{Lie}(T)$, the level set of H_1 is preserved by H_2 , so $\{H_1 = c_1, \dots, H_r = c_r\}$ is connected when nonempty. By an argument of Atiyah, the image is a convex polytope, which is the convex hull of the images of the fixed points.

For instance, if T is the maximal torus in G and M is the (co)adjoint orbit, then this picks out the diagonal elements of a matrix in O . For $G = \text{SU}(3)$, the image looks like below:

Figure 3.7: Polytope associated to $\text{SU}(3)$

3.6 Root Systems

Let G be a compact Lie group and T be a maximal torus. Then we know that

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \neq 0} \mathfrak{g}_{\alpha}.$$

We will prove that the α form a reduced root system and that $\dim \mathfrak{g}_{\alpha} = 1$. We know that the roots sit inside the root lattice, which sits inside the character lattice of T , which is $\mathbb{Z}^r$.

Definition 3.6.1. A finite collection of vectors in Euclidean space forms a *root system* if

1. For all roots α, β , the vector

$$r_{\alpha}(\beta) = \beta - 2 \frac{(\alpha, \beta)}{(\alpha, \alpha)} \alpha$$

is also a root. Here $(-, -)$ is the metric on $\mathfrak{t}$.

2. For all roots α, β , the quantity $2\frac{(\alpha, \beta)}{(\alpha, \alpha)}$ is an integer.

A root system is called *reduced* if for roots α, β , the equation $\beta = c\alpha$ implies $c = \pm 1$.

Note that there is an unreduced root system $BC_r = \{e_i - e_j, e_i, 2e_i\}_{i,j \leq r}$, where the e_i form an orthonormal basis of Euclidean space. However, this does not appear in the context of Lie theory.

We will now classify root systems of rank 1. The only such root systems are $A_1 = \{\alpha, -\alpha\}$ and $BC_1 = \{\pm\alpha, \pm 2\alpha\}$. The root system A_1 is the root system of $G = SU(2)$ with $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}(2)$. Here, we have

$$\mathfrak{g}_{\alpha} = \mathbb{C} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \mathfrak{g}_{-\alpha} = \mathbb{C} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

To classify root systems of rank 2, we note that

$$\frac{4(\alpha, \beta)^2}{(\alpha, \alpha)(\beta, \beta)} \in \mathbb{Z}.$$

This implies that $4 \cos^2 \angle(\alpha, \beta)$ is an integer. The simplest is called $A_1 + A_1$, with the two roots perpendicular. Next, the angle could be $\pi/3$, and the root system is now A_2 , which is drawn below.

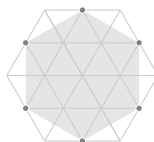


Figure 3.8: The A_2 root system

This is the root system of $SU(3)$. The next root system, in the case where the angle is $\pi/4$ is the root system B_2 and C_2 , and it looks like



Figure 3.9: The B_2 root system

B_r is the root system of $SO(2r+1)$ and C_r is the root system of $Sp(2r)$. Next, we have the root system G_2 , which is drawn below:

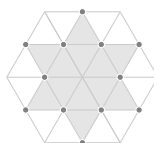


Figure 3.10: The G_2 root system

In general, all finite, or even discrete, groups generated by reflections of $\mathbb{R}^n$ can be explicitly classified. This restricts to the classification of crystallographic groups (which preserve a lattice),

and inside this we can classify reduced root systems. This turns out to be the same as classifying compact Lie groups and complex reductive Lie groups.

The first step is for every α , find $r_\alpha \in W$. Last time, we showed that $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{C}h_\alpha \subset \mathfrak{t}_\mathbb{C}$. Then we know $(h_\alpha, h) = c\alpha(h)$. Now we can normalize this by $\alpha(h_\alpha) = 2$. In fact, we will see that the h_α form the dual root system in $\mathfrak{t}$. Next, for any α , we can choose $e_\alpha \in \mathfrak{g}_\alpha$, which defines an action of $\mathfrak{sl}(2)$ on $\mathfrak{g}_\mathbb{C}$. In particular, we have

$$[e_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\beta+\alpha} \quad [e_{-\alpha}, \mathfrak{g}_\beta] = \mathfrak{g}_{\beta-\alpha}.$$

Therefore a root β corresponds to the submodule $\bigoplus \mathfrak{g}_{\beta+n\alpha}$, and in particular, we have the submodule

$$\bigoplus_{n \neq 0} \mathfrak{g}_{n\alpha} \oplus \mathbb{C}h_\alpha$$

and so h_α is the unique vector of zero weight. Now we know the representation theory of $SL(2)$, where the highest weights are integers. Also note that we only have even weights and a unique vector of weight 0, and thus the representation above is irreducible. However, by construction, it contains $\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha} \oplus \mathbb{C}h_\alpha$ as a subrepresentation, so we must have

$$\mathbb{C}h_\alpha \oplus \bigoplus_{n \neq 0} \mathfrak{g}_{n\alpha} = \mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha} \oplus \mathbb{C}h_\alpha.$$

This proves that $\dim \mathfrak{g}_\alpha = 1$ and that the α are reduced. Also, we know that in the diagram

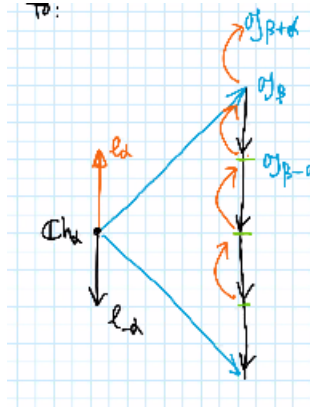


Figure 3.11: Root String

the representation $\bigoplus_n \mathfrak{g}_{\beta+n\alpha}$ is irreducible. These are reversed by $r_\alpha = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and thus $\text{Ad}(r_\alpha)$ acts by reflection on $\mathfrak{t}$ and $\mathfrak{t}^*$. This implies that

$$2 \frac{(\alpha, \beta)}{(\alpha, \alpha)} \in \mathbb{Z}.$$

Our next goal is to show that W is generated by the r_α . Before we do this, we will discuss the group generated by reflections. For example, if we consider an equilateral triangle, we can find a discrete group generated by reflections.

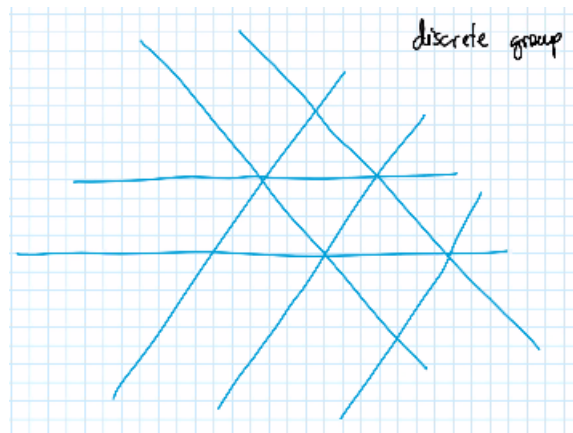


Figure 3.12: Reflection hyperplanes

We can consider the connected components of the complement, and these are called *polytopes*. These can be studied in $\mathbb{R}^n$, S^n , $\mathbb{H}^n$, and other spaces. Each of these is a finite intersection of half-spaces and is called a *polyhedron*. Then if $r_1, \dots, r_n$ are the reflections in the facets of Δ , they generate a group Γ . Now we can rephrase any word in the generators as a path, so for example, we have

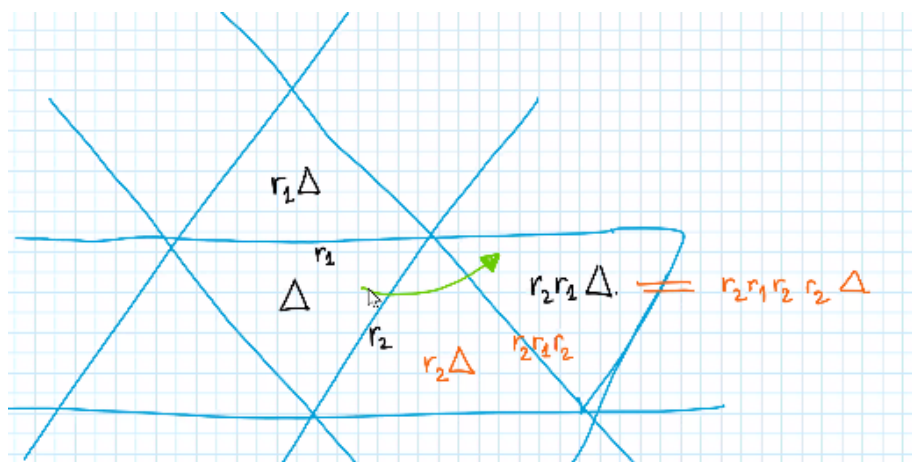


Figure 3.13: Words in the generators

Similarly, we can obtain $r_1r_2r_3\Delta$ by the procedure

$$\Delta \rightarrow r_1\Delta \rightarrow r_1r_2r_1(r_1\Delta) \rightarrow r_1r_2r_3r_2r_1(r_1r_2r_1(r_1\Delta)).$$

Therefore, the group $\langle r_1, r_2, \dots, r_n \rangle$ acts transitively on the set of chambers. In particular, any reflection r_α is one of these reflections. Now there are some obvious relations on the r_i :

1. Clearly $r_i^2 = 1$.
2. If two hyperplanes r_1, r_2 have angle π/m between them, then we have $(r_1r_2)^m = 1$.

Theorem 3.6.2. 1. This is the complete list of relations among the r_i

2. Δ is a fundamental domain for Γ .

The idea of the proof is to remove intersections of hyperplanes of codimension at least 3. This is still simply connected, but it has a map

$$\left\{ \bigsqcup \Gamma \times \Delta \right\} \rightarrow \mathbb{R}^n \setminus \{\text{codimension} \geq 3 \text{ intersections}\}$$

which is a covering, and therefore it must be an isomorphism.

To say some more about W , if W is generated by relations, recall that W corresponds to critical points on G/T with index $\ell(w)$. Then we will see that for a vector ξ , the relation $\alpha > 0$ if and only if $\alpha(\xi) > 0$ gives us a chamber for $\langle r_\alpha \rangle$. Therefore by the action of $\langle r_\alpha \rangle$ we can change the index to be 0, but there is only one cell of index 0 because G/T is connected.

Recall that if Γ is a discrete group generated by reflections, a *chamber* Δ is a connected component of the complement of the reflecting hyperplanes.

Theorem 3.6.3. 1. For a chamber Δ , $\bar{\Delta}$ is a fundamental domain for Γ .

2. Γ is in bijection with the set of chambers.

3. Γ is generated by the reflections r_i in the walls of Δ with the relations $r_i^2 = 1$ and $(r_i r_j)^m = 1$ whenever the walls r_i, r_j have angle π/m .

Last time, we proved that the r_i generate Γ . Next, by the following picture, we see that words in generators correspond to paths in the set of chambers:

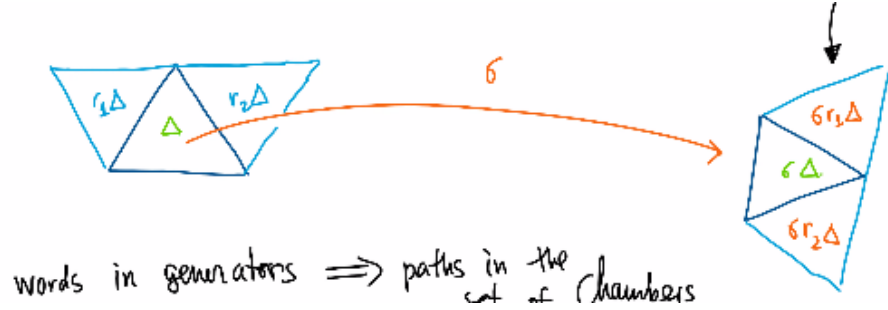


Figure 3.14: Words.

Proof. We will view words in the generators of a group as the Cayley graph. This is the graph with vertices the group elements and edges $\gamma_1 \xrightarrow{r_i} \gamma_2$ if $\gamma_2 = r_i \gamma_1$. Because $r_i^2 = 1$, we don't need these orientations. Now let

$$\tilde{\Gamma} = \langle r_i \rangle / (r_i^2 = 1, (r_i, r_j)^{m_{ij}} = 1) \rightarrow \Gamma \rightarrow \{\text{chambers}\}$$

be given by $\gamma \mapsto \gamma\Delta$. This takes the Cayley graph of $\tilde{\Gamma}$ to the adjacency graph of chambers. We will prove this map is an isomorphism. This will imply that $\tilde{\Gamma} = \Gamma$ and that Γ acts freely on the set of chambers. In fact, we will beef it up to a covering map of something simply-connected.

Consider the complement of the codimension 3 strata in $\mathbb{R}^n$. Also consider the set

$$\tilde{\Gamma} \times (\Delta^3 \setminus \{\text{vertices}\}) /$$

where we identify $\gamma \times \Delta$ with $\gamma r_i \times \Delta$. This is clearly a local isomorphism near points on codimension 2 strata, and therefore is an isomorphism because the target is simply connected. $\square$

Example 3.6.4. Consider the *triangle groups* $\Gamma = \langle r_1, r_2, r_3 \rangle / (r_i^2 = 1, (r_i r_j)^{m_{ij}} = 1)$. This is the reflection group of either $\mathbb{R}^2, S^2, \mathbb{H}^2$ generated by a triangle with angles $\frac{\pi}{m_{12}}, \frac{\pi}{m_{13}}, \frac{\pi}{m_{23}}$. Here, the cases are

$$\frac{1}{m_{12}} + \frac{1}{m_{23}} + \frac{1}{m_{13}} = \begin{cases} > 1 & \Delta \in S^2 \\ 1 & \Delta \in \mathbb{R}^2 \\ < 1 & \Delta \in \mathbb{H}^2. \end{cases}$$

For example, consider the triangle generated by three perpendicular great circles on the sphere. Each angle is $\frac{\pi}{2}$. In $\mathbb{R}^2$, we can consider the tiling of the plane by usual equilateral triangles, and in the hyperbolic plane, the fundamental domain of the $SL(2, \mathbb{Z})$ action gives us a $(2, 3, \infty)$ triangle.

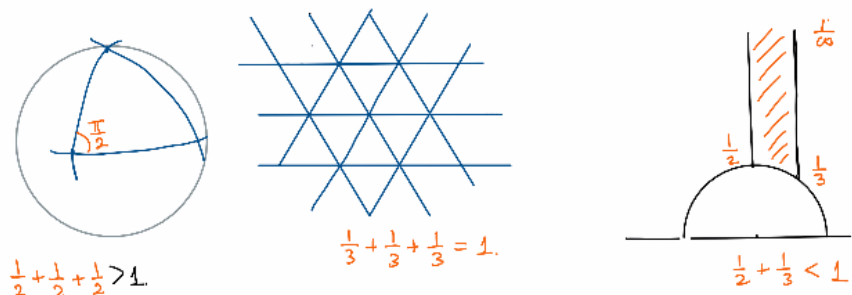


Figure 3.15: Triangles

In these cases, we know Γ is

- Finite if $\sum \frac{1}{m_{ij}} > 1$.
- Infinite of growth like $\mathbb{R}^2$ for $\sum \frac{1}{m_{ij}} = 1$.
- Infinite of exponential growth if $\sum \frac{1}{m_{ij}} < 1$.

Recall that the Weyl group $W = N(T)/T$ is a finite group.

Theorem 3.6.5. W is generated by the r_α for all roots α .

Proof. Identify W with the critical points of $(\xi, -)$ on the orbit of ξ in $\mathfrak{g}$ for a generic $\xi \in \text{Lie}(T)$. We need to prove that there exists no $w \in W$ such that $w\xi$ is in the same chamber as ξ . On the critical point $w\xi$, we will compute the index of the function $(\xi, -)$. We know this is the number of roots α such that $\alpha(\xi) > 0$ and $\alpha(w\xi) < 0$, which is the same as the number of hyperplanes separating ξ and $w\xi$. This also equals the number of reflecting hyperplanes separating Δ from $w\Delta$, which is the length of w .

Therefore, if there exists $w\xi$ in the same chamber as ξ , then there is more than one maximum of $(\xi, -)$. However, because there are no 1-cells, the orbit of ξ is disconnected, which is impossible because G is connected. $\square$

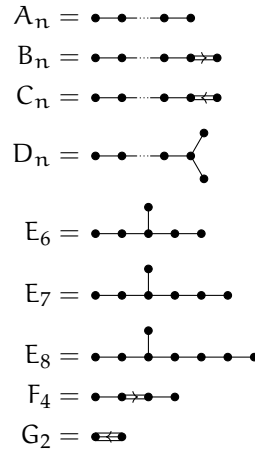
Now we have the following correspondence:

Compact group $\rightarrow$ root system $\rightarrow$ finite reflection group $\subset$ discrete reflection groups.

Discrete reflection groups in $\mathbb{R}^n$ can be classified, so from this we can infer a classification of root systems. Next, the map from compact groups to root systems is an isomorphism, so compact groups (or complex reductive groups) can be classified. The conclusion is

- There are four infinite series $SU(n+1), SO(2n+1), Sp(2n), SO(2n)$ denoted by $A_n (n \geq 2), B_n (n \geq 2), C_n (n \geq 3), D_n (n \geq 4)$ corresponding to the different Dynkin diagrams.
- There are several simple exceptional groups: G_2, E_6, E_7, E_8, F_4 . As a representation of $\mathfrak{sl}_3$, we have $\mathfrak{g}_2 = \mathfrak{sl}_3 \oplus \mathbb{C}^3 \oplus (\mathbb{C}^3)^\vee$. Note that $\mathfrak{sl}_3$ has an outer automorphism, which is $A \mapsto -A^T$. From this decomposition and the root system, one can reconstruct the remaining brackets.

Now a *Dynkin diagram* is a diagram with vertices the r_i and an edge between r_i and r_j when $m_{ij} = 3$ and no edge when $m_{ij} = 2$. When $m_{ij} = 4$, we have to choose which edge is longer, so the Dynkin diagrams are



Bonus: E_6, E_7, E_8 and algebraic surfaces

Recall that algebraic curves (or compact Riemann surfaces) come in three forms: $\mathbb{P}^1$ (genus 0, Fano), elliptic curves ($g = 1$, Calabi-Yau), and curves of genus $g > 1$ (general type). Each of these exhibits very different behavior when we consider meromorphic differentials ω . Then if we consider the canonical class $K_C := (\omega)$. For any two meromorphic differentials ω, ω' , we see that ω/ω' is a rational function, so all forms define the same element of $\text{Pic } C$. Also, we can compute using Riemann-Roch that $\deg K_C = 2g - 2$.

For example, on $\mathbb{P}^1$, the form dx has a double pole at infinity. Then if we consider a rational function $f: C \rightarrow \mathbb{P}^1$, we can consider the form $f^* dx$ and then compute the degree using Riemann-Hurwitz. This implies that when $g = 0$, K_C has negative degree, when $g = 1$, the degree is zero, and when $g > 0$, the degree is positive.

For general algebraic varieties X , we can consider the canonical divisor K_X associated to a meromorphic top form. Now let S be a smooth projective surface in $\mathbb{P}^N$.

Example 4.0.1. The simplest such S is $\mathbb{P}^2$, and we can consider the form $dx \wedge dy$ on $C^2 \subset \mathbb{P}^2$, and this will have a triple pole at the line at infinity. Alternatively, if we consider the form $\frac{dx}{x} \wedge \frac{dy}{y}$, this has poles on the toric boundary. This implies that $K_{\mathbb{P}^2} = -3H$, which is the hyperplane class.

Example 4.0.2. Let $S_d \subset \mathbb{P}^3$ be a surface cut out by a polynomial of degree d . When $d = 2$, we have a quadric surface, which look like this:



Figure 4.1: Quadric Surface

This has two families of lines parameterized by $\mathbb{P}^1$, and in fact, $S_2 \simeq \mathbb{P}^1 \times \mathbb{P}^1$. Note that this is *birational* to $\mathbb{P}^2$ but is not isomorphic. Projection from a point $x_0 \in S_2$ is a map $S_2 \dashrightarrow \mathbb{P}^2$ and is an isomorphism outside of the projection point and the two lines through it. In fact, to resolve the

base point of this map, we need to blow up x_0 . The two rulings through x_0 are contracted. Here, blowup replaces a point by a line representing all tangent directions through it.

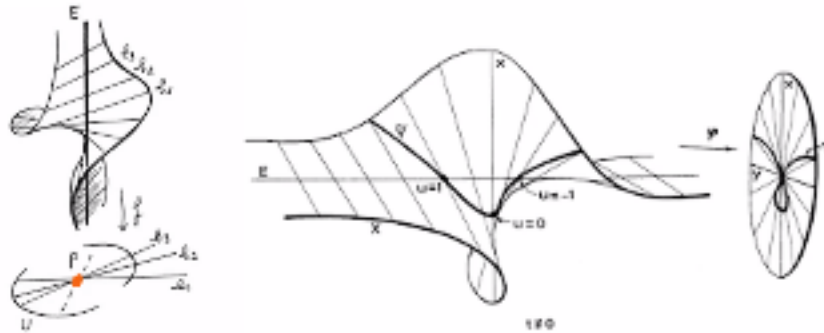


Figure 4.2: Blowup of a smooth point on a surface

In the toric picture, blowup and blow down are represented by the following:

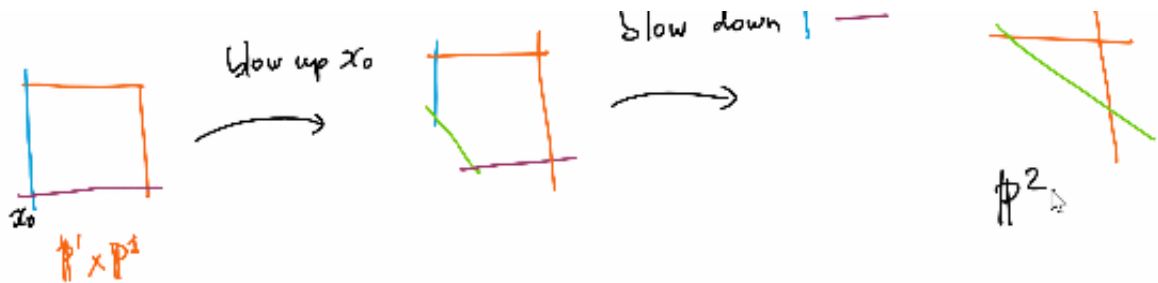


Figure 4.3: Toric blowup and blowdown

Then we obtain $K_{\mathbb{P}^1 \times \mathbb{P}^1} = -2\text{pt} \times \mathbb{P}^1 - 2\mathbb{P}^1 \times \text{pt} = -2H$ because the hyperplane section is the sum of two $\mathbb{P}^1$.

In general, we can compute the canonical class of a hypersurface $\{P = 0\} = Y \subset X$ using the adjunction formula. We can write $\omega_Y = \frac{\omega_X}{dP}$ and thus we have

$$K_Y = (K_X + Y)|_Y.$$

For example, if $Y = S_d \subset X = \mathbb{P}^3$, we see that $K_{S_d} = (d - 4)H$. This tells us that

- When $d = 1, 2, 3$, the surface S_d is a *del Pezzo surface*.
- When $d = 4$, we have a K3 surface.
- When $d > 4$, we have a surface of general type.

When $d = 3$, the surface is a cubic surface and looks like this:

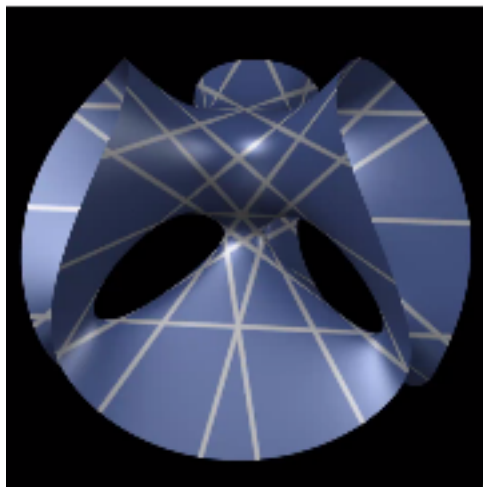


Figure 4.4: Cubic Surface

Over $\mathbb{C}$, every smooth cubic surface has 27 lines. We have $K_S = -H$ and thus $-K_S = H$, which is *ample*. Recall that a divisor is ample if some multiple of it defines an embedding into $\mathbb{P}^N$. Now there is a map $\text{Pic } S \rightarrow \text{NS}(S)$ from the Picard group to the Neron-Severi group, which is the group of divisors modulo numerical equivalence. Here, we need the intersection form $D_1 \cdot D_2 \in \mathbb{Z}$, which is additive, invariant under deformation, and counts intersection points with multiplicity when $D_1 \cap D_2$ is discrete and D_1, D_2 are *effective*. Here are some basic results:

1. The Neron-Severi group of a surface is a free abelian group $\mathbb{Z}^\rho$, and ρ is called the *Picard rank*.
2. The signature of the intersection form is $(1, \rho - 1)$.

Inside the cone where $C^2 = 0$, we can consider the *ample cone* of ample divisors. The dual of this is the closure of the effective cone. There are effective divisors with negative self-intersection, for example any line on a cubic surface.

Now we may prove that the cubic surface is rational. To do this, we choose two skew lines L, L' . Now for any points $x, x' \in L, L'$, we consider the line through x, x' in $\mathbb{P}^3$, and there is a third intersection point. This defines a birational map $S \dashrightarrow \mathbb{P}^2$, and in fact S is the blowup of $\mathbb{P}^2$ at six points. When we blow up, we obtain $K_S = K_{\mathbb{P}^2} + \sum_{i=1}^6 E_i$, and thus we have $K^2 = 9 - 6 = 3$. Thus $-K$ is very ample and defined an embedding into $\mathbb{P}^3$. This tells us that

$$\text{Pic } S = \mathbb{Z}H \oplus \bigoplus_{i=1}^6 \mathbb{Z}E_i.$$

If $-K_S$ is in the closure of the ample cone and $C \subset S$ is a curve, we will consider how small C^2 can be. By the adjunction formula, we have

$$g(C) = \frac{C^2 + C \cdot K}{2} + 1 \geq 0.$$

Therefore, we have $C^2 \geq -2 - K \cdot C$. Thus we have $-K \cdot C \geq 0$ and this is positive when $-K$ is ample. This implies that $C^2 \geq -2$, and this happens when $K \cdot C = 0$. If $C^2 = -1$, then C is a line and can be contracted by Castelnuovo's criterion, and when $C^2 = -2$, we can contract this to an A_1 singularity (or $\frac{1}{2}(1, 1)$). If we consider the perpendicular of $-K$, then all (-1) -curves are:

- $C = E_i$;
- $H - E_i - E_j$;
- $2H - \sum_{k=1}^5 E_{i_k}$.

This gives us the 27 lines on a cubic surface: the six exceptionals, the strict transforms of the 15 lines between two of the blown up points, and the 6 conics passing through five of the six points. The (-2) -curves are:

- $C = E_i - E_j$;
- $C = H - \sum_{i=1}^3 E_i$;
- $C = 2H - \sum_{i=1}^6 E_i$;
- $C = 3H - 2E_1 - \sum_{i=2}^8 E_i$.

The first corresponds to blowing up a point twice, the second corresponds to blowing up three points on a line, and the third corresponds to blowing up six points on a conic. The last one corresponds to a cubic with a double point at p_1 and seven more points. The last ones correspond to the roots of E_6, E_7, E_8 . To see this, if we consider a singular S_0 and nearby nonsingular S_t , we have a vanishing cycle in $H^2(S_t)$, which is a root. If we consider the monodromy, this is exactly the reflection in this root.

This defines a map from $H^2(S, \mathbb{Z})$ to the family of all nonsingular cubic surfaces. Unfortunately, we did not get to the punchline, which was to actually choose a set of roots in this cohomology.